



9TH IEEE INTERNATIONAL TEST CONFERENCE INDIA 2025
JULY 20-22, 2025 RADISSON BLU, BENGALURU, INDIA

PROCEEDINGS

Our Sponsors

Platinum



SIEMENS

TESSOLVE
A HERO ELECTRONIX VENTURE

Gold

cādence®

Google

synopsys®

Qualcomm

Bronze

ANORA

DeFT
Semiconductors

 **TEXAS INSTRUMENTS**

SANDISK™

Silver

 **MARVELL™**

TERADYNE

ADVANTEST

Message from the Technical Program Chairs

Welcome to the 2025 IEEE International Test Conference India (ITC-India), the 9th in a series of events dedicated to exploring emerging trends, innovative research, and scalable solutions for the challenging problems of electronic test of devices, boards, and systems.

The two-day technical program features exciting sessions encompassing research and applications across a wide range of test-related topics. In addition to traditional test topics such as design-for-test (DFT), reliability, functional test, and validation, this year's program introduces new and emerging areas. These include the role of AI in test—covering both the testing of artificial intelligence (AI) hardware and the application of AI in test methodologies—as well as 3D IC and chiplet testing, and discussions on addressing silent data errors. The program also features a compelling panel discussion on "India's Semiconductor Test Ecosystem: Building for Scale and Innovation".

ITC-India continues to attract many submissions not just from India but also internationally. For ITC-India 2025, the Technical Program Committee (TPC) received a staggering 398 paper submissions with author affiliations from 17 participating countries. The TPC, comprising around 50 members from both industry and academia, thoroughly reviewed the submissions, with each submission receiving at least 3 reviews. The final program was selected by the TPC to include 30 high-quality technical papers covering the entire gamut of Test and DFT related topics. The program also covers testing issues related to emerging technologies and applications.

As is the norm, the ITC-India program also includes the traditional Test Reality Check, Industry Sessions, and Academia Research Tracks. This year, ITC-India has invited six distinguished speakers to present six keynotes which will pose intriguing questions on test issues in AI hardware, silicon lifecycle management, and smart testing strategies focused on test cost optimization and quality improvement. We also have an exciting distinguished address series planned on the topic of "Transformation of DFT: From being a necessary evil to strategic differentiator", where experts from the industry will provide their valuable perspectives and insights on this critical subject.

The TPC expresses profound gratitude to all the invited speakers, who are leaders in their respective fields, for generously sharing their vision and experience with the attendees. The ITC-India technical program is the result of the volunteer efforts of more than 30 dedicated researchers and professionals. The TPC members have made particularly significant contributions to ensure a high-quality program. The TPC chairs extend their sincere thanks for these invaluable efforts. The overarching goal of the ITC-India committees is to deliver a program that addresses upcoming and future test challenges, meeting your evolving needs, as well as the contemporary challenges faced by the test community.

ITC-India is your conference. We earnestly encourage your active participation. The TPC chairs sincerely hope that your lively interaction will make ITC-India 2025 interesting, valuable, and fun!

TPC Chairs,



Kavitha Shankar
Nordic Semiconductor, UK



Dr. Subhadip Kundu
Qualcomm, India



Wilson Pradeep
Tenstorrent, USA

Message from the General Chairs

Dear Colleagues,

It is our great honor and pleasure to welcome you to the **9th IEEE International Test Conference India (ITC India 2025)**, taking place from **July 20–22, 2025**, in the dynamic city of Bengaluru, India. As General Chairs representing both industry and academia, we are proud to bring together a diverse and vibrant community of professionals, researchers, and students to explore the latest innovations and challenges in the field of semiconductor testing.

Our dedicated organizing committee has prepared a dynamic program that highlights the evolving and critical nature of the semiconductor testing domain. This year's conference is our most extensive yet, with over 800 attendees participating in the event. The technical program features **30 full papers, 20 posters, 10 invited talks, 6 keynote addresses, and 4 distinguished addresses**, offering a comprehensive view of the state of the art in test and validation. We are also hosting a high-impact panel discussion on "**India's Semiconductor Test Ecosystem: Building for Scale and Innovation**", which brings together leaders from across the value chain to discuss how India can strengthen its global role in semiconductor testing.

A hallmark of ITC India is its commitment to bridging academic research and industrial practice. The Academia Research Track showcases pioneering work from leading institutions, while the Industry Test Challenge and Hackathon provide hands-on opportunities to solve real-world problems. With four parallel technical tracks, the conference ensures a rich and diverse learning experience for all attendees. Since its inception in 2017, the ITC India conference has become a cornerstone for networking and knowledge sharing. We conduct numerous workshops across India and proudly offer over 100 fellowships for academics to connect with the industry.

We are also excited to present 10 tutorials across four parallel tracks, covering topics such as power device testing, silicon lifecycle management, and test security. Additionally, the Learning Corner Series focused on **ATMP (Assembly, Test, Marking, and Packaging)** and **OSAT (Outsourced Semiconductor Assembly and Test)** offers practical insights into the evolving back-end ecosystem—an area of growing strategic importance.

We thank you for being part of ITC India 2025 and look forward to your active participation. Whether you are here to present, learn, or connect, we hope this conference inspires you and contributes meaningfully to your journey in the world of semiconductor test.

Warm regards,



Sivanantham S
VIT Vellore



Gaurav Bhargava
Qualcomm, India

ITC INDIA 2025 COMMITTEE

General Chairs

Sivanantham S
VIT Vellore

Gaurav Bhargava
Qualcomm, India

Technical Program Co-Chairs (TPC)

Kavitha Shankar
Nordic Semiconductor, UK

Subhadip Kundu
Qualcomm, India

Wilson Pradeep
Tenstorrent, USA

Tutorials Co-Chairs

Bharath Nandakumar
Cadence, India

Lakshmanan Balasubramanian
Texas Instruments, India

Mehala Balasundaram
Synopsys, India

Abhishek Chaudhary
AMD, India

Industry Test Challenges Co-Chairs

Anuj Gupta
Cadence Design Systems

Nikhil Sudhakaran
Marvell, India

Achin Grover
Google

Industry Sessions Co-Chairs

Shamitha Rao
Synopsys

Sandeep Jain
Siemens EDA

Academia Research Track Co-Chairs (ART)

Prof. Usha Mehta
Nirma University, Ahmedabad

Leela Krishna Thota
Synopsys, India

Prof. Binod Kumar
IIT, Jodhpur

Test Reality Check Co-Chairs (TRC)

Dr. Rajit Karmakar
AMD, India

Veejaye Panayadian
Sandisk, India

Panel Chair

Kamlesh Pandey
Krivya Semicon, India

Posters Co-Chairs

Pranjal Giri
Texas Instruments

Rama Sireesha Arisetti
Google, India

Publication Co-Chairs

Prof. R. Jayagowri
BMSCE, Bangalore

Prof. Sakthivel Ramachandran
VIT, Vellore

ITC INDIA 2025 COMMITTEE

Fellowship Co-Chairs

Prakash Talawar
Western Digital, India

Prof. Saravanan P
PSG Tech, Coimbatore

Piyushkumar Chaniyara
Texas Instruments, India

Finance Chair

Krishnan Sreenivasan
AB Innovative

Communication/Website Co-Chairs

Nithin Bharadwaj
Microchip, UK

Abhirami R
Samsung Semiconductor India Research

Registration Chair

Leela Krishna Thota
Synopsys

Dharani Srinivasan
Qualcomm, India

Vikram
AMD, India

Marketing & Logistics Chair

Veeresh Shetty
Mentor – A Siemens Business

Samuel Dorairaj
Intel

TPC Advisory Committee

Prof. Alex Orailoglu
University of California, USA

Prof. James Chien-Mo Li
National Taiwan University, Taiwan

Dr. C.P. Ravikumar
India

R. D. (Shawn) Blanton
Carnegie Mellon University, USA

Advisory Committee

Navin Bishnoi
Marvell Semiconductors, India

Prasad Mantri
Eximietas, India

Dr. Yervant Zorian
Synopsys

Prof. Adit Singh
Auburn University

Prof. Virendra Singh
IIT Bombay, India

Prof. Susmita Sur-Kolay
ISI, India

Scott Davidson
Retired Computer Scientist, USA

Dr. Rubin Parekhji
Texas Instruments, India

Ron Press
Siemens EDA, USA

Prof. Ujjwal Guin
Auburn University, USA

Jais Abraham
Qualcomm, India

Sameer Chillarige
Cadence, India

Venkata Rangam Totakura
Infineon Technologies, India

AGENDA

DAY 0

REGISTRATIONS				
08:00am - 09:15am	HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA
	TRACKS	Track 1 (a): Analog and Mixed-Signal Test and Failure Analysis Mehala Balasundaram	Track 2 (a): Silicon to Systems Test Abhishek Chaudhary	Track 3 (a): Test Security Bharath Nandakumar
09:15am - 10:45am	SESSION CHAIR	Advanced Power MOSFET Testing: Methodologies, Challenges, and Best Practices Manoj Pachaiyan, Kalyana Sundaram Chandran and Senthilkumar Dhamodharan (Caliber-Interconnects)	Silicon Lifecycle Management: Pivotal Component in Extension of Silicon to Systems Leela Krishna Thota, Ravi Teja Vemuri and Veeramani Rajalingam (Synopsys)	Silent Data Corruption: A Designer's View Ankush Srivastava (Qualcomm)
10:45pm - 11:15pm		Tea/Coffee Break		
11:15am - 12:45pm		Electrical Failure Analysis for Low Power Mixed-Signal Integrated Circuit Failure Debug - Tools & Techniques Lakshmanan Balasubramanian, Muruges Prashanth Subramaniam, Srinivas V V, Sourabh Gupta	UTAG – Lifeline of Modern Era SoC DFT Divyank Mittal, Sagar Kumar and Vaibhav Mishra (Cadence)	Scan based Testing and Test Security Challenges: Scan and Security can Live Together Jayagowri R (BMS College of Engineering)
12:45pm - 01:45pm		Lunch Break		
	HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA
	TRACKS	Track 1 (b): Recent Advances in Test Lakshmanan Balasubramanian	Track 2 (b): 3D Integration & Test Mehala Balasundaram	Track 3 (b): Memory Test Bharath Nandakumar
01:45 pm - 03:15 pm	SESSION CHAIR	A Review of Recent Advances in Test: Cell Aware Test, Timing Aware Test and System Level Tests - Part I Adit Singh (Auburn University)	Demystifying DFT: Basics of VLSI Testing and Evolution of DFT Abhishek Chaudhary and Arun Vasudevan (AMD)	Advances in Memory Design and Testing: Insights from Academia and Industry - Part I Sakthivel Ramachandram (VIT Vellore) Sanjith Sleeba, Rahul Sahu and Subhadip Kundu (Qualcomm)
03:15pm - 03:45pm		Tea/Coffee Break		
03:45pm - 05:15pm		A Review of Recent Advances in Test: Cell Aware Test, Timing Aware Test and System Level Tests - Part II Adit Singh (Auburn University)	Demystifying DFT for 3D: Test Techniques and Repair Strategy for Multi-Die and 3D Devices Lee Harrison (Siemens EDA)	Advances in Memory Design and Testing: Insights from Academia and Industry - Part II Sanjith Sleeba, Rahul Sahu and Subhadip Kundu (Qualcomm)
5:15 pm - 6:15 pm				
		Invite only Session (Industry Test Challenge)		

AGENDA

DAY 1

08:00am - 09:00am	REGISTRATIONS			
HALL NAME	GRAND VICTORIA			
09:00am - 09:30am	Conference Inauguration General Co-Chair, ITC India 2025 (Opening Remarks, Conference Highlights, Lamp Lighting by Chief Guests)			
09:30am - 10:00am	Keynote Address Transformative Design for Test Technologies for Silicon Lifecycle Management Janusz Rajski (Siemens EDA, USA)			
10:00am - 10:30am	Keynote Address Future of AI Hardware Enabled by Advanced Packaging Raja Swaminathan (AMD, USA)			
10:30am - 11:00am	Keynote Address Enabling efficient prototyping for Fabless design houses Rajesh V (Tessolve)			
11:00am - 11:05am	Exhibit Zone Inauguration			
11:05am - 11:30am	Tea/Coffee Break			
HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA	BRAIN BOX
SESSIONS	Paper Session 1: AI in Test: Detect, Predict, Compensate Manivannan Ethiraj	Paper Session 2: ATE Intelligence: Speed Meets Precision Abhishek Chaudhary	Paper Session 3: Accelerated Chiplet Testing: Interconnects to IUTAG Rajesh Kumar Tiwari	Academic Research Track (ART) Session 1 Leela Krishna Thota
SESSION CHAIR	1.1: Anomaly Ranking of LPDDR4 Shmoo Results using Machine Learning Vinodh J Rakesh and Chaitanya Kumar Reddy Mallu (Infineon Technologies, India) 1.2: Machine Learning-Based High Vmin Prediction for Enhanced Pre-Silicon Reliability Sudheer Anumala, Satya Ramoji Chikkam (Intel, India); Yu Tin Cheong, Xin Rou Chong and Mohd Azwann Zulkefle (Intel, Malaysia) *1.3: A Machine Learning-based Structural Built-in Self-Test for Lifetime Frequency Drift Ritik Gupta (IISc Bangalore, India); Aswin R, Rubin Parekhji (Texas Instruments, India) and Gaurab Banerjee (IISc Bangalore, India)	2.1: Test Time Reduction in Current Sense Amplifiers via Optimized Current Range Across Wafer Touchdowns and Gain Reordering on V93K SMT8 Hemanthkumar V (Infineon Technologies, India) 2.2: Accelerating Time to Market of Semiconductor IC's using Smart ATE PCB Routing System B Lokapriya, A Dyaneswaran, R Karthika, S Lokendran and Dharmodharan Senthilkumar (Caliber Interconnects, India) 2.3: Accelerating ATE Pattern Verification & Debug using Emulation Platform - Enabling First-pass Success on Silicon Kumar Gaurav Singh, Praveen Kumar Ravichandran, Saurabh Karkun, Karthik Rajakumar, Yogeshwaran Shanmugam and Pratibha Gupta (Texas Instruments, India)	3.1: Test Selection and Scheduling for 3D-Stacked Chiplet-based Designs Ravikumar C P (Vinyana Tech, India) and Shiva Shankar B (IIT Dharwad, India) 3.2: Effective 3D IC Interconnect Test Using Defect Distance and Bump Placement Location Anshuman Chandra (Siemens DIS, USA); Barbara Dzialowska, Marta Stepieniewska (Siemens DIS, Canada); Chi-Chun Yang (Siemens DIS, Taiwan); and Jonathan Gaudet (Siemens DIS, Canada) *3.3: Implementation and Optimization of High-Bandwidth IUTAG for Enhanced Test Time Reduction in Complex SoCs Sandipan Sharma, Nikita Naresh (Google, India); Wilson Pradeep (Google, USA); Karthick Prabhu (Siemens DISW, Poland); and Olga Przybycz (Siemens DISW, India)	ART 1.1: LLMCoV: A Methodology for LLM-aided DFT Coverage Improvement in Open-Source Designs Manisha Kumari, Subharthi Roy (IIT Jodhpur, India); Masahiro Fujita (University of Tokyo, Japan); and Binod Kumar (IIT Jodhpur, India) ART 1.2: Optimal Test Point Insertion for Enhanced Small Delay Defect Coverage Faraz Aatif, Prathiba Muthukrishnan and Sivanantham Sathasivam (VIT Vellore, India) ART 1.3: Assertion-Driven Formal Verification for Custom RISC-V Core Sree Sankar E, Sakshi Thakur and Binod Kumar (IIT Jodhpur, India) ART 1.4: Enhanced Power Estimation of Digital VLSI Circuits using Machine Learning Maniyar Mohammed Saqlain and Sakthivel Ramachandran (VIT Vellore, India) ART 1.5: Implementation of Memory Built-in Self-Test Amol G. Patil, Suhas B Shirol and Vikas K R (KLE Technological University, India)
01:00pm - 02:00pm	Lunch Break			

AGENDA

DAY 1

HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA	BRAIN BOX
SESSIONS	Paper Session 4: Aging & Escapes: The Silent Threat Nikita Nareish	Paper Session 5: From Design to Debug: Silicon Validation Rahul Anand	Paper Session 6: Innovative Test for Flexible & Mixed-Signal Ics Kiran Rajmohan	Academic Research Track (ART) Session 2 Prof. Binod Kumar
SESSION CHAIR	4.1: From Structural Test Escapes to Silent Data Errors: A preliminary analysis Paolo Bernardi, Francesco Angione (Politecnico di Torino, Italy); and Arani Sinha (Intel, USA) 4.2: Lite-CMOS Transition Fault Model for Legacy Design Yi Sun (NXP Semiconductors, USA), Qiong Wang (NXP Semiconductors, China), Saidapet Ramesh (NXP Semiconductors, USA) and Cunyu Shi (NXP Semiconductors, China) *4.3: Assessing Long-Term Reliability through Accelerated Aging Stress Tests on a On-chip Quality and Reliability Platform : A Band Gap Reference case study Shiv Shankar Gupta, Javed GS and Rahul Sharma (Intel, India)	*5.1: VTEST: FPGA-Based SoC Validation Framework Srikanth Neelam, Sreekanth Arati (VConnecTech Systems Pvt. Ltd, India); and Amalin Prince A (BITS Pilani, India) 5.2 Novel approach for implementation and validation of scan dump debug methodology Savan Bhatelia and Mahesh Bendre (Intel, India) 5.3: Novel Post Silicon Validation Strategies for the New Era of UCle Chiplets Bhuvana Chandar, Anirudha Rao, Santhosh Kumar S. Meghana, Bukkapathnam, Deepak Baliga (Intel, India); and Chad E Beach (Intel, USA)	6.1: An Integrated Environment for Simulation of Mixed Signal Tests using Open-Source Tools Gursimran Singh, Irawati Thete, Pammy Thakur and Rajesh Jabade (Texas Instruments, Bangalore) 6.2: A Smart Framework for Cost-Effective Characterization of Touch-Sensing Controller Paul Ngoy (Infineon Technologies, USA); Abhishek Ganesh Poojary, Jagadish Raju K and Keerthan Rai (Infineon Technologies, India) 6.3: Automated BIST Technique for Interconnect Characterization and Reliability Enhancement in Flexible Electronic Circuits Harshad Singh, Kirti Srivastava and Oppili Prasad L (IIT-BHU, India)	Special Address by Dr. Sreejit Chakravarty (Ampere Computing, USA) - 30 mins General Chairs & Fellowship Students Dialogue - 30 mins VLSI Test Coverage Optimization Hackathon Introduction & Talks by Top-3 Teams (30mins) Titans - KLE Technological University, Hubli Flipflop fixers - BMS College of Engineering, Bengaluru Quantum Bytes - Indian Institute of Information Technology, Guwahati Honorable Mention: Impro_DFT, Einfachips (An Arrow Company) pvt. Ltd.
02:00pm - 03:30pm				
03:30pm - 04:00pm	Tea/Coffee Break			
HALL NAME	GRAND VICTORIA			
04:00pm - 04:25pm	Distinguished Address The Standard for Test - IEEE 1838 for 3D-IC Adam Cron (Synopsys)			
04:25pm - 05:30pm	Panel Discussion India's Semiconductor Test Ecosystem: Building for Scale and Innovation Panelists: Ruchir Dixit (Siemens), Paresh Bharkhada (Teradyne), Anand Muthaiah (Tessolve), Senthil (Caliber Interconnects) Moderator: Venkata Ranganathakura (Infineon)			
05:30pm - 05:45pm	Closing Remarks General Co-Chairs, ITC India 2025			
	Poster Session (BANQUET ENTRANCE LOBBY)			

AGENDA

DAY 2

08:30am - 09:00am	REGISTRATIONS			
HALL NAME	GRAND VICTORIA			
09:00am - 09:30am	Welcome / Day 1 Summary General Co-Chair, ITC India 2025			
09:30am - 10:00am	Keynote Address The Right Testing Strategy Can Save Designs Nitza Basoco (Teradyne)			
10:00am - 10:30am	Keynote Address Rethinking Silicon Test to reduce DPPM and SDC Dr. Sreejit Chakravarty, IEEE Fellow and Distinguished Engineer (Ampere Computing, USA)			
10:30am - 11:00am	Keynote Address From Silicon to Solutions: The Societal Impact of Semiconductor Leadership Suresh Babu K (Caliber Interconnect Solutions, India)			
11:00am - 11:30am	Tea/Coffee Break			
HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA	BRAIN BOX
SESSIONS	Industry Session 1 Shamitha Rao	Paper Session 7: Smart SoC Testing: Access, Memory, Reliability Sudhir Raut	Paper Session 8: Secure and Efficient SoC Test Infrastructure Rajesh Khurana	Test Reality Check (TRC) Rajit Karmakar
11:30am - 01:00pm	Advanced Yield Analysis Capabilities for the Hyper-scaler Era Kort Longenbach (Marvell)	*7.1: Exploring Tesseract Two Pin Serial Port for MBIST, LBIST and ATPG for a low pin count SoC Ratheesh Thekke Veetil, Soumyajyoti Paul, Prakash Kumar, Anvesh Gadi (Analog Device, India) ; Ajay Purushotham and Viद्या Neerkundar (Siemens EDA, India)	8.1: Hybrid Bscan and Self-Test IOs to Reduce IO Test Time Mathangi Raghuraman, Sujith Thomas, Sharmila Jayakumar and Ramakrishnan Subramanian (Sandisk, India)	TRC 1.1: IDDq Vector Quality (FTRV) Improvement on Mature Technodes through Prime Time Power Extension (PTPX) Simulation at Pre-Silicon Marijune Samaniego, Hariharan Krishnamurthy (Qualcomm, Singapore) ; Sudheer Yadapalli, Shilpa Bhardwal, Tanu Pruthi and Maharshi Pandya (Qualcomm, India)
	Serving the Era of Complexity through Continuous Innovations Raj Singam (Advantest)	7.2: Enhanced RTL Test Point Insertion Flow with Improved Functional Block Testability Nikita Nareish, Naushad Ali (Google, India); Wilson Pradeep(Google, USA); Oussama Laouamri (Siemens USA); Surya Phanindra Chatla, Karthick Prabhu B (Siemens India); Niranjan Mukherjee (Siemens USA)	8.2: Concurrent ATE testing of MSIP PHYs in SOC via Scan Streaming Kriti Sundar Das, Paras Jain (Google LLC, India); and Harini Kulkarni (AMD, India)	TRC 1.2: Incremental Memory Repair Techniques with Synopsys SMS for Multi-hierarchy Server Architecture Ankit Bansal (Synopsys, India); and Ajay Chauhan (Marvell, India)
	Emerging challenges and new approaches in Validation of Mixed signal SoCs Pradeep Nair (Texas Instruments)	7.3: Enhancing Memory Reliability: Integrating Error Correction Codes with Memory Built-In Self-Test for Effective Validation; Luc Romain (Siemens, Canada); Paul-Patrick Nordmann (Siemens, Germany); Albert Au (Siemens, Canada); Wei Zou, Harshitha Kodali, Martin Keim and Lori Schramm (Siemens, USA)	8.3: FuseLock: Dynamic Taint-Based Scan Chain Protection with Fuse-Guarded Counters Niranjana Ithal, Gautham Varma Kanumuru, Lasya Chidanand Hegde, Tanish Shet and Sudeendra Kumar K (PES University, India)	TRC 1.3: Automated and Verifiable Security Exclusion Management for Scan-Based Debug: The SDEX Methodology Krunal Siddhapathak and Harsharaj Ellur (Google, India)
01:00pm - 02:00pm	Lunch Break			
				TRC 1.4: Redefining Low Power Scan Testing: A Novel Asymmetric Clock Gating Archdeep Singh, Navaneeth R and Vishal Diwan (Texas Instruments, India)
				TRC 1.5: Breaking the Drop: Innovative Methods for IR Drop

AGENDA

DAY 2

HALL NAME	GRAND VICTORIA 1	GRAND VICTORIA 2	ARABICA & ROBUSTA	BRAIN BOX
SESSIONS	Industry Session 2 Sandeep Jain	Paper Session 9: Test Cost Optimization: Scan and LBIST Advances Anirudh Kadiyala	Paper Session 10: Secure Resilient Fabrics: Designing for Reliability and Unclonability Prof. Binod Kumar	Learning Corner Series Veejaye Panayadiyan
SESSION CHAIR	Strategic Test Optimization in the Era of Flash Memory Scaling Prakash Talawar (Sandisk)	9.1: A Left Shifted Approach for Low Shift Power ATPG Decompression Hillol Maity, Hanumant Tuntoni, Sreenu Kakunuri (Synopsys, India); Peter Wohl (Synopsys, USA); and Parthajit Bhattacharya (Synopsys, India)	10.1: EFH - An Efficient Fault-Tolerant Routing Methodology for 2D Mesh NoCs Biswajit Bhowmik, Girish K K, Akot Jayesh Raju and Rohit Chakraborty (IIT Karnataka, India)	Learning Corner Series - ATPM/OSAT Fundamentals Speakers: ATE in Action: Teradyne on Semiconductor Manufacturing Test & Value Creation Paresh Bharkhada (Teradyne)
02:00pm - 03:30pm	Anora Labs : Scope of Functional Testing in India Dr. Pramod Variyam (Anora Labs)	9.2: Optimized Grouping for Packetized Scan Solutions Bhavika Ranjeet Kumar, Aalhad Deshpande, Rama Sireesha Arisetti and Maheedhar Jalasutram (Google, India)	*10.2: Design and Analysis of Complementary Cascade Current Mirror based Physically Unclonable Function using 65nm Predictive Technology Model Shreeji Shah and Rajesh Thakker (Gujarat Technological University, India)	Scaling with Confidence: Seamless Navigation for High-Volume Semiconductor Testing, Larry Liow (Advantest Singapore)
	From Black Box to Glass Box: A Novel XAI Approach with GNNs for Test and Debug Dr. Shiva Kumar Kotikalapudi (DeFT Semiconductor)	9.3: Tester time savings for LBIST diagnostics using enhanced version of On-chip MISR signature comparison with self-test and bypass Jaidev Shenoy, Kelly Ockunzi (Marvel Technology Inc.) and Virendra Singh (IIT Bombay, India)	10.3 Invited Talk: Testing Systems-in-Package based on Chiplests : Opportunities and Challenges Wilson Pradeep (Tenstorrent, USA)	Production Infrastructure and ATE Environment - The Fundamentals, Challenges of Existing Flows and Future Trends Gaurav Mittal and Rajesh Jabade (Texas Instruments)
03:30pm - 04:00pm	Tea/Coffee Break			
HALL NAME	GRAND VICTORIA			Poster Session (BANQUET ENTRANCE LOBBY)
04:00pm - 05:15pm	Distinguished Address Series Transformation of DFT: From being a necessary evil to strategic differentiator Jais Abraham (Qualcomm) ; Pradeep Nagaraj (Cadence) ; Srinivas Vooka (Google)			
05:15pm - 05:45pm	Awards Ceremony and Valedictory session General Co-Chair, ITC India 2025			

KEYNOTE ADDRESS

KEYNOTE 1

Transformative Design for Test Technologies for Silicon Lifecycle Management

Janusz Rajski, Siemens EDA, USA



Bio— During his tenure at Mentor Graphics and Siemens, Janusz has built a strong R&D organization with a focus on innovative DFT and SLM technologies. The team has developed several revolutionary products: TestKompress, Cell-Aware Test, and Streaming Scan Network. He has published more than 280 IEEE research papers and is a co-inventor of 130 US and international patents. A Lifetime Fellow of the IEEE, he holds a Ph.D. degree in electrical engineering as well as an honorary doctorate from the Poznań University of Technology. In 2003, he was awarded the prestigious title of “Professor of Science” by the President of Poland. In 2009 he received the Stephen Swerling Innovation Award from Mentor Graphics “for his breakthrough innovation, TestKompress, and his many contributions to revitalizing Mentor’s DFT business to its current position as the #1 test business in EDA”. In 2018, Janusz received the Siemens Inventor of the Year Lifetime Achievement Award “in recognition of his outstanding achievements which have led to valuable inventions in the field of DFT of integrated circuits”. In 2022 he received the Siemens Inventor of the Year Award for co-inventing the Streaming Scan Network.

KEYNOTE 2

Future of AI Hardware Enabled by Advanced Packaging

Raja Swaminathan, AMD



Bio— Dr. Raja Swaminathan is the Corporate Vice President of Packaging at AMD, spearheading the development of AMD’s advanced packaging and heterogeneous integration roadmap. With a distinguished career spanning roles at Intel, Apple, and now AMD, Dr. Swaminathan’s expertise in design-technology co-optimization and dedication to optimizing power, performance, area, and cost (PPAC) have led to significant technological advancements such as EMIB, Apple’s Mx packages, 3D V-Cache, and 3.5D architectures for AI accelerators. Dr. Swaminathan holds a PhD from Carnegie Mellon University and an undergraduate degree from IIT Madras. With over 100 patents and more than 40 published papers to their name, Dr. Swaminathan was recently recognized as an IEEE Fellow and serves as a technical advisor to multiple startups. His unwavering commitment to heterogeneous integration continues to drive the boundaries of silicon technology. He also shares his insights on life and leadership, drawing lessons from the semiconductor industry, on his LinkedIn profile.

KEYNOTE ADDRESS

KEYNOTE 3

The Right Testing Strategy Can Save Designs

Nitza Basoco, Teradyne



Bio—Nitza Basoco is the Technology and Market Strategist in the Semiconductor Test Division at Teradyne, focusing on start-ups, emerging companies and hyperscalers. With over 24 years of experience in the semiconductor industry, Nitza is a proven leader with deep technical expertise in bringing emerging technologies to market. Prior to Teradyne, Nitza was the Vice President of Business Development at proteanTecs, responsible for their partnership strategy and value-add ecosystem growth. Before that she was Vice President of Operations of Synaptics and held various leadership positions at MaxLinear. Her time as a principal test development engineer (TDE) at Broadcom provided vital experience in semiconductor test.

Nitza holds a Master of Engineering and a Bachelor of Science degree in Electrical Engineering from MIT and an Executive Master of Business Administration (EMBA) from Quantic School of Business and Technology. She is an active volunteer with the Global Semiconductor Alliance Women's Leadership Initiative (GSA WLI), a mentor in the GSA Women in Hardware program (WHM) and on the National Academy Foundation (NAF) Advisory Board of the Compton Unified School District.

KEYNOTE 4

Rethinking Silicon Test to reduce DPPM and SDC

Dr. Sreejit Chakravarty, IEEE Fellow and Distinguished Engineer, Ampere Computing.



Bio—Dr. Sreejit Chakravarty is an IEEE Fellow, a highly recognized Researcher, Inventor, and a Distinguished Engineering Leader, with extensive industry and academic experience.

He is currently a Distinguished Engineer at Ampere Computing, Santa Clara, CA, USA where he drives strategic initiatives for product quality. Prior to this he had over 25 years of industry experience as a Principal Engineer with Intel Corporation and Distinguished Engineer at LSI and AVAGO (now Broadcom). He started his career in academia as an Associate professor of Computer Science, at The State University of New York at Buffalo, where his work was funded by multiple National Science Foundation Grants.

KEYNOTE ADDRESS

He has architected innovative solutions across the entire silicon life cycle spanning Silicon Quality and Reliability (RAS, Functional Safety and Silent Data Errors); and subsequently drove them from concept to product intercept.

He has published 1 book, authored 145+ IEEE papers and has 23 issued US patents. He has served in various capacity at numerous IEEE conferences and delivered multiple keynote addresses, the latest being at the IEEE Asina Test Symposium, 2023. He has mentored research at several universities like Princeton, USC, UIUC, etc. For his professional work he has been recognized as an IEEE Fellow and SUNY Distinguished Alumni. He currently chairs the IEEE P3405 Work Group on Chip-let Interconnect Test and Repair, which aims to standardize the test and repair of chip-let interconnects which will lay the foundation to realize the chip-let revolution.

KEYNOTE 5

From Silicon to Solutions: The Societal Impact of Semiconductor Leadership

**Suresh Babu K, Managing Director,
Caliber Interconnect Solutions Pvt. Ltd,
Coimbatore**



Bio—Suresh began his entrepreneurial journey at the age of 32 in the year 2000, driven by his passion for business and software. Remarkably, he had already developed his first test hardware design IPs 25 years earlier, during a time when modern EDA tools were still in their infancy. Capitalizing on his deep technical expertise, he became a trusted vendor, offering design and test development services to major semiconductor and ATE (Automatic Test Equipment) companies.

Throughout his career, Suresh successfully built and led high-performing teams specializing in test development, test hardware design, packaging, and high-speed digital/RF simulation. His interests extended beyond semiconductor electronics, branching into power systems, medical devices, and robotics. Under his leadership, these multidisciplinary teams delivered cutting-edge solutions to clients in the automotive and railway sectors. Suresh holds an engineering degree and a postgraduate degree in Consultancy Management from BITS Pilani, further complementing his technical and managerial acumen.

KEYNOTE ADDRESS

KEYNOTE 6

**Enabling efficient prototyping for
Fabless design houses**

**Rajesh V, Senior VP – Test and Product
Engineering, Tessolve**



Bio— Over 25 years of experience with a solid track record of driving revenue growth, leading organizations, and enhancing strategic partnerships. As the P&L owner for global Test Engineering services, specialize in a range of technologies, including Digital, Analog, Mobile, Automotive, Cloud, and AI. These efforts have successfully expanded the Tessolve Test Business Unit to include over 1500+ engineers across divisions such as Bench Characterization Engineering, Test and Product Engineering, Quality, and Automation. We have set up the Test labs with high-volume manufacturing equipment in San Jose and Austin, which are equipped with advanced Automated Test Equipment (ATEs), Probers, and Handlers, setting new standards in industry capabilities.

TUTORIAL

TRACK 1A

Analog and Mixed-Signal Test and Failure Analysis Session Chair : Mehala Balasundaram

Advanced Power MOSFET Testing: Methodologies, Challenges, and Best Practices

09:15am - 10:45am

GRAND VICTORIA 1



Manoj Pachaiyan
Team Lead of IC Testing,
Caliber Interconnect

Manoj Pachaiyan holds a Master's degree in Power Electronics from Anna University and has over 10 years of professional experience, including more than 5 years in IC testing. Currently serving as the Team Lead of IC Testing at Caliber Interconnect, he specializes in post-silicon validation of Power Management ICs, mixed-signal testing, and high-speed interface blocks. With a strong foundation from his earlier academic career as an Assistant Professor in Analog Electronics, Manoj has developed deep expertise in both ATE and bench-level testing. His work includes advanced validation methodologies, automation using Python and LabVIEW, and hands-on experience with tools like oscilloscopes, SMUs, and high-speed test instruments.



Kalyana Sundaram Chandran
Test Lead,
Caliber Interconnect

Kalyana Sundaram Chandran possessing 18 years of experience and he is currently working as a Test Lead at Caliber Interconnect Solutions, where he focuses on testing digital and mixed-signal ICs using various tester platforms. He specializes in research and development, creating advanced testing techniques and methodologies to drive product innovation and quality. His notable contributions in R&D also include progress in biomedical VLSI signal processing and programming in Verilog and VHDL. He has an impressive academic record with 25 international journal publications and over 20 papers presented at international conferences. He holds two patents and two copyrights. He earned his Ph.D. in Biomedical VLSI Signal Processing from Anna University, Chennai. Outside of work, he enjoys reading historical novels and traveling, which fuel his passion for creative thinking.



Senthilkumar Dhamodharan
Director,
Semiconductor Test Engineering,
Caliber Interconnect

Senthilkumar Dhamodharan currently serves as Director of Semiconductor Test Engineering at Caliber, where he leads the Test Engineering Division, responsible for overseeing the testing of Digital/Mixed Signal/RF products on ATE/Bench platforms. He is a proactive advocate for driving the development of intellectual property (IP) to stimulate innovations in silicon testing. Within his role, he shoulders the responsibility of mapping out the strategic direction for Test Engineering, wielding his influence across a wide array of areas, including the adept leadership of teams in Physical Design and DFT. Senthil takes immense pride in his well-established track record of assembling high-performance testing teams, a formidable force comprising over 200 skilled test engineers. Their primary mission is to consistently deliver outstanding value to clients on a global scale. He holds a Master's degree from Anna University in VLSI Design and boasts an impressive 18+ year tenure working with global technology giants such as Qualcomm, AMD and NXP. Driven by an authentic passion for technology, he derives immense joy from engaging with individuals and embarking on journeys to explore diverse destinations around the world.

TUTORIAL

Abstract: Power MOSFETs play a crucial role in modern power electronics, enabling efficient energy management in applications ranging from consumer electronics to industrial automation. Evolving from early planar structures to advanced Super Junction and Wide Band Gap (SiC/GaN) technologies, MOSFET design has continually progressed to meet the demands of higher switching speeds, improved efficiency, and greater power handling. These technological advancements, while beneficial, have also introduced new challenges in testing—such as high-speed switching effects, parasitic influences, and increasingly complex measurement requirements. This tutorial aims to provide a comprehensive discussion on MOSFET fundamentals, classification, and key electrical parameters, followed by advanced testing methodologies. Topics include gate charge characterization, dynamic $R_{DS(on)}$ measurement, high-voltage testing complexities, and parasitic capacitance measurement. Practical challenges such as feasibility analysis, resource selection, and PCB design considerations will also be addressed. The session will explore the integration of precision instruments, such as LCR meters and PAM/CAM modules, into Automated Test Equipment (ATE) to improve test accuracy. Furthermore, best practices for MOSFET testing, including calibration techniques, reducing parasitic effects, ensuring safe high-voltage testing, and effective thermal management, will be discussed. Special emphasis will be placed on UIL testing and MOSFET avalanche ruggedness evaluation for reliability assessments. Attendees will gain a deep understanding of MOSFET characterization techniques, equipping them to enhance test performance and accuracy in engineering and mass production environments.

Electrical Failure Analysis for Low Power Mixed-Signal Integrated Circuit Failure Debug – Tools and Techniques

11:15am - 12:45pm

GRAND VICTORIA 1



Lakshmanan Balasubramanian
Technologist & Senior Member
Technical Staff,
Texas Instruments

Lakshmanan Balasubramanian is an Technologist and Senior Member Technical Staff at Texas Instruments (TI), focusing on AMS SoC design integration and verification. With over 25 years of experience, he specializes in analog and embedded PM design, low-power digital and AMS design/verification, analog DFT/test, and silicon debug. He has authored 145+ technical papers, holds 11 patents, and received 3 TI Manufacturing Incentive Awards. Lakshmanan has won over 10 best paper/presentation awards and has delivered numerous keynotes, invited talks, tutorials, and trainings across industry and academia. He actively contributes to standards development through IEEE, Accellera, and Si2, particularly in AMS and low-power domains. His key roles include:

- IEEE 1801 (UPF) – currently Secretary
- Contributions to IEEE 1800 (SV-AMS), 1687.2, 2427, 2416
- Member of Accellera SV-AMS, SC-AMS, UVM-AMS
- Active in Si2 CDC and SiG on Machine Learning

A Senior Member of IEEE, he is also affiliated with IET, ACM, IPRA, PRI, ACCS, VSI, and is a Chartered Engineer (UK). He is recognized within TI and the wider industry as a go-to expert for AMS/LP design, verification, and debug, and frequently consults across TI business units. As TI's Industrial Liaison to the Semiconductor Research Corporation (SRC), he drives global research collaborations with institutions like CEG/AU, Arizona State, Iowa State, IIT Kharagpur, IIT Roorkee, BITS Hyderabad, Oxford, and UC Berkeley. He is a two-time recipient of the Mahboob Khan Outstanding Liaison Award (2016, 2023). Lakshmanan has served as a TPC member and session/track chair for leading conferences including DAC, ETS, VTS, VLSID, VDAT, ITC India, and TIITC, and is a frequent reviewer for journals and events such as IEEE TCAD, D&T, ACM TODAES, DAC, DATE, and more. He holds a B.Sc. in Physics from University of Madras, a B.Tech in Electronics from MIT, Anna University, an M.Tech in Electronic Design from IISc, and completed a predoctoral program at IMEC/KU Leuven.

TUTORIAL



Murugesh Prashanth Subramaniam
Analog Design Manager &
Member Group Technical Staff,
Texas Instruments

Murugesh Prashanth Subramaniam is an Analog design manager & Member Group Technical Staff, responsible for analog RF design integration & verification. He has 20+ years industry experience and expertise in power management (PM) with significant expertise in all aspects of PM including buck and boost DCDC converters, complete system level PM care-about, real time clocking, wake-up and power domain management of mixed-signal SoC with integrated RF and Wi-Fi radio. Has led several highly complex customer/RMA debug activities across businesses in TI. Published 7 conference papers, 6 patents. He holds B.E. Electronics and Comm. Eng., from VIT, 2001; and M.S. Elect. Eng., Boise State Univ., Idaho, in 2003.



Srinivas V V
Design Manager & Senior
Member
Technical Staff
Texas Instruments

Srinivas V V is a Design Manager & Senior Member Technical Staff, responsible for WiFi radio development. He has 20+ year industry experience with expertise in design of PM LDOs, DC-DC converters, baseband and RF circuits. He has published 10 conference and journal papers & 9 patents. He holds B Tech, Elect. Eng., IIT Madras, 2001., and M.S, IIT Madras, 2006.



Sourabh Gupta
Member Group Technical Staff,
Texas Instruments

Sourabh Gupta has been with TI for 21 years, an MGTS, has worked on system, validation, test and application engineering of high speed analog to digital converters (10 MSPS to 3 GSPS) for about 16 years. In recent part of his career, he has focused on designing high speed link (for up to 56 Gbps serial data transmission) from die to PCB and its measurement in lab. He has also been designing various microwave circuits within substrate of IC package for TI's bleeding edge AFE technologies. He has been instrumental in developing signal processing algorithms for finding faults in automotive high-speed signal chains. He is a widely sought-after expert within TI in RF, microwave, EMC/EMI/ESD domain.

Abstract: This tutorial will introduce various electrical failure analysis techniques and their application to integrated circuit failure debug and root-cause analysis through industrial case studies through illustrative examples of simple but common failure signatures due to circuit design, system integration, low power, reliability issues and defects. The focus will be on techniques, tips and tricks towards failure analysis of analog mixed-signal and low power design & integration from a practicing design & test engineers' perspective.

TUTORIAL

TRACK 1B

Recent Advances in Test

Session Chair : Lakshmanan Balasubramanian

A Review of Recent Advances in Test: Cell Aware Test, Timing Aware Test and System Level Tests

PART 1: 01:45 pm - 03:15 pm

GRAND VICTORIA 1

PART 2: 03:45pm - 05:15 pm



Dr. Adit Singh
Chair Professor, Electrical
and Computer
Engineering
Auburn University

Dr. Adit D. Singh is Godbold Endowed Chair Professor of Electrical and Computer Engineering at Auburn University. He earlier served on the faculties of the University of Massachusetts in Amherst, and Virginia Tech in Blacksburg, and has held visiting positions at the University of Tokyo, Japan, the Universities of Freiburg and Potsdam in Germany, the Indian Institute of Technology, and as a Fulbright scholar at the University Polytechnic of Catalonia in Barcelona, Spain. His technical interests span all aspects of VLSI technology, particularly integrated circuit test and reliability. He has published over three hundred research papers and holds international patents that have been licensed to industry. He has served as a consultant to several semiconductor and EDA companies, including as an expert witness for major patent litigation cases. He has had leadership roles as General Chair/Co-Chair/Program Chair for dozens of international VLSI design and test conferences. He served two terms (2007-11) as Chair of the IEEE Test Technology Technical Council (TTTC), and (2011-15) on the Board of Governors of the IEEE Council on Design Automation (CEDA). Singh received his B.Tech from IIT Kanpur, and the M.S. and Ph.D. from Virginia Tech, all in Electrical Engineering. He is a Life Fellow of IEEE.

Abstract: This tutorial offers a comprehensive discussion of recent advances in digital IC testing. We begin with a concise review of classical stuck-at (SA) and transition delay fault (TDF) models, along with commonly used scan and test compression design-for-test (DFT) methodologies, to establish a foundational context. The main discussion commences with Cell-Aware Test (CAT). We explore its motivation, test generation methodology, and its application using both single-cycle and two-pattern tests. Drawing on industrial data, this section also examines the resulting increase in test set size for CAT patterns, the enhanced defect coverage—including unique defects identified by CAT over SA and TDF—and silicon results demonstrating DPPM improvement. Next, we present Timing-Aware Tests. This section provides a detailed review of scan timing test, LOC and LOS test applications, and a thorough exploration of transition and path delay fault models. We explain how timing-aware tests (including timing-aware cell-aware tests) are designed to target small delay defects, discussing their effectiveness and limitations. The final section addresses the growing necessity of functional System-Level Tests (SLTs). Due to unacceptable levels of test escapes from DFT-based scan structural tests, SLTs have become an indispensable final test screen for high-performance processors and smartphone ICs. We explain the fundamental differences between SLT testers and traditional ATEs and the diverse failures targeted by each test system.

TUTORIAL

TRACK 2A

Silicon to Systems Test

Session Chair : Abhishek Chaudhary

Silicon Lifecycle Management: Pivotal Component in Extension of Silicon to Systems

09:15am - 10:45am

GRAND VICTORIA 2



Leela Krishna Thota
Synopsys, India

Leela is currently working in Synopsys as Manager, Hardware Analytics Team. Current work focus is on Silicon Life cycle Management (SLM), related subsystem development driving its initiatives. Integration of ON-chip sensors, designing for analytics, enabling predictive maintenance and performance optimization by combining traditional DFT techniques with modern SLM methodologies are of prime focus. Overall VLSI experience is 13 years in a collateral production role at various companies that includes the exciting work involved with Samsung SSIR (Foundry Design Services), Altran & SiCon Design Technologies Pvt. Ltd. Besides DFT, also gained VLSI board design knowledge in Wipro. Received various accolades for the exceptional work at different clients like Microsemi (now Microchip) & Qualcomm, DRDO. Post graduated on VLSI Design from SRM University & was awarded a gold medal.



Veeramani Rajalingam
Synopsys, India

Veeramani is a seasoned VLSI Engineer with over six years of experience in Physical Design. He has significantly contributed to high-impact projects such as HBM3IO, ARM CPU, and Gaming SoCs. Currently, he serves as a Staff Engineer at Synopsys Inc., where he oversees SLM IP implementation activities for clients in India. Veeramani holds an MTech degree in VLSI from the National Institute of Technology Goa. Beyond his professional role, he is passionate about teaching engineering students VLSI-Backend concepts and Mathematics, emphasizing practical applications.



Ravi Teja Vemuri
Synopsys, India

Ravi Teja Vemuri is a semiconductor test engineering professional with over 12 years of experience in developing and validating test content across leading ATE platforms. At Synopsys, he is working on post-silicon bring-up, driving high-temperature operating life (HTOL) activities, and test methodology development for Silicon Lifecycle Management (SLM) IPs. His expertise spans test enablement, pattern conversion automation, production test optimization and data analytics. Ravi has presented multiple technical papers at industry events and internal conferences, focusing on pre- and post-silicon test flows, scan compression, and pattern validation strategies. He holds an M.Tech in Microelectronics from BITS Pilani and is actively involved in patent development related to test time optimization techniques.

TUTORIAL

Abstract: In an era defined by complex chip architectures, ever-shrinking technology nodes and very demanding applications, Silicon Lifecycle Management (SLM) has become a foundational strategy for optimizing performance, reliability, and efficiency across the lifespan of a semiconductor device. SLM provides a comprehensive strategy for managing the data and processes associated with the maintenance and service of targeted components throughout their entire lifecycle. By integrating SLM with predictive maintenance, cybersecurity measures, and adherence to industry standards, manufacturers can ensure that maintenance activities are not only timely but also aligned with the overall vehicle service strategy. SLM products are built on a foundation of improving silicon health and operational metrics at every phase of the device lifecycle by enriched in-chip observability, analytics and integrated automation. Monitors enable deep insights from silicon to system. Meaningful data is gathered at every opportunity for continuous analysis and actionable feedback, improving design efficiency and quality while also predicting in-field chip degradation or failure. SLM products continuously monitors critical silicon metrics like voltage, temperature, margins, and health, enhancing the reliability and performance of ECUs, CPUs, GPUs, and other architectures through real-time edge and cloud analytics. This enables the assessment of silicon aging and reliability, providing valuable insights into both systemic and random defects and facilitating predictive maintenance to extend silicon lifecycle and reduce costs.

IJTAG – Lifeline of Modern Era SoC DFT

11:15am - 12:45pm

GRAND VICTORIA 2



Divyank Mittal
Cadence India

Divyank Mittal is Principal Software Engineer at Cadence Design Systems, India . He is part of the MODUS Test Solutions team and has work experience of 12 years. His main contributions are in the development of software EDA solutions for DFT chips having a standardized Access and Control mechanism for Embedded Instruments . He holds multiple US patents for inventing innovative methodologies to solve complex problems and challenges encountered by a pattern retargeting EDA tool while operating the access network to control and observe Embedded IPs. He had also written and presented a ITC paper last year 2024 on "Optimize Test Pattern Count via Efficient Operation of IEEE 1687 SIBs" which was later on published for IEEE publications. He is also good at many sports like badminton, table tennis, swimming, football etc. His hobbies include cooking national and international cuisines, travelling and playing e-games.



Sagar Kumar
Cadence India

Sagar works as Principal Product Engineer in Modus DFT Software Solution group at Cadence Design Systems, Noida. His 7+ years of experience at Cadence involves working on multiple areas of DFT mainly including IJTAG, Hierarchical Test, Boundary Scan Verification and 3DIC. He holds two patents in his name. He received his M. Tech in VLSI Systems from VIT, Vellore

TUTORIAL



Vaibhav Mishra
Cadence India

Vaibhav Mishra is a Lead Software Engineer at Cadence Design Systems working primarily in iJTAG automation and Modus Diagnostics solution. Vaibhav has educational experience in electronics and computer science and has been working in EDA domain for 5 years. In his free time, he likes to unwind by playing computer games and reading fictional literature.

Abstract: With the ever-increasing complexities in ASIC SOCs iJTAG has become the lifeline of modern era DFT for enhanced testing and debugging of embedded instruments such as wide varieties of sensors, monitors, controllers etc. at the chip, board, and system levels. This tutorial will initially cover the basics of iJTAG in detail which includes what is iJTAG, why is it needed and what are the constructs of iJTAG and then gradually move on to advanced iJTAG concepts like SIBs and eTAPs and their hierarchical network operability. This tutorial also covers the wider applications of iJTAG in the DFT domain specifically in controlling the LBIST and MBIST controllers, hierarchical test setup and iJTAG with 3D ICs, its benefits, value propositions and how can EDA tools perform automation in iJTAG where this tutorial will cover in detail the automated network insertion/updating/tracing mechanisms and automated pattern retargeting to reduce engineers' efforts and optimize test time significantly.

TRACK 2B

3D Integration & Test

Session Chair : Mehala Balasundaram

Demystifying DFT: Basics of VLSI Testing and Evolution of DFT

01:45 pm - 03:15 pm

GRAND VICTORIA 2



Abhishek Chaudhary
Principal Member of
Technical Staff,
Server DFX
AMD India

Abhishek Chaudhary works as Principal member of technical staff in the Server SOC DFT team at AMD. He is an experienced DFT Engineer with 18 years of experience in the semiconductor industry. Prior to joining AMD, Abhishek has worked at Texas Instruments, Rambus and Freescale Semiconductor in different capacities, primarily in the area of DFT and test.

Abhishek has a passion for continued education and is an active volunteer in TTTC events and has presented on DFT topics at various external events and conferences. Abhishek holds a Master's degree from IIT Delhi and Bachelors from Visvesvaraya National Institute of Technology, Nagpur.

TUTORIAL



Arun Vasudevan
Principal Member
of Technical Staff,
Server DFX
AMD India

Arun Vasudevan works as Principal member of technical staff in the Server SOC DFT team at AMD. He has 18+years of experience in DFT. Arun has also worked at ARM and Intel in different capacities mostly in the area of DFT. Arun holds a bachelor's degree in electronics from MS Ramaiah institute of technology, Bangalore.

Abstract: This tutorial is intended to give a sneak peek to the audience to the world of VLSI testing and the various challenges that make this field an interesting and lively area to work on. We will introduce the various types of testing methods, the different stages of testing in the silicon lifecycle, and discuss concepts of scan testing, MBIST and DFT standards. The session will start with an introduction to VLSI testing and end with setting the stage for 2.5D IC testing that will be covered in the next tutorial of this track.

Demystifying DFT for 3D: Test Techniques and Repair Strategy for Multi-Die and 3D Devices

03:45pm - 05:15pm

GRAND VICTORIA 2



Lee Harrison
Director, Product Marketing
Siemens Tessent Division

Lee Harrison is Director, Product Marketing, with Siemens Tessent Division. He has over 20 years of industry experience with Siemens Tessent DFT products and has been involved in the specification of new test features and methodologies for Siemens customers, delivering high quality DFT solutions. With a focus on safety and security, Lee is working to ensure that current and future DFT requirements of Siemens's Automotive customers are understood and met. Lee Received his BEng in MicroElectronic Engineering from Brunel University London in 1996. Lee presents regularly at industry conferences such as DAC, ITC, VTS, ETS, DATE etc.

Abstract: Multi-die products and in particular 3D products will have significant impact on how Design-For-Test (DFT) needs to be implemented, executed, and monitored throughout the life cycle of the product. There are many, many pitfalls for incorrect and incomplete DFT that will result in high test costs in both die area and test time, yield loss, or high shipped DPPM rates. None of this is desirable. On the other hand, DFT done right will add value to the product and company. In this session, you will learn how DFT is evolving from classical "2D" designs to provide value-add solutions addressing the challenges of huge, multi-die products. Topics of discussion include so called "Known-good-die", test of the multi-chip assembly, especially to detect defects between the dies, their repair and diagnosis, related test standards like IEEE 1838 and P1838a, IEEE P3405, IEEE 1687, interfaces like PCIe, USB, and UCIe, and many other topics. Addressing the increased test quality requirements while at the same time keeping the test cost in check is the underlying theme throughout. At the end of the session, you will understand that for multi-die, 3D products, DFT can no longer be an afterthought, but needs to become an upfront planned aspect supporting the lifecycle of the product, DFT needs to become part of the entire 3D ecosystem.

TUTORIAL

TRACK 3A

Test Security

Session Chair : Bharath Nandakumar

Silent Data Corruption: A Designer's View

09:15am - 10:45am

ARABICA & ROBUSTA



Ankush Srivastava
Principle Engineer,
Qualcomm, India

Ankush Srivastava is currently involved in enabling Industry leading DFT methodologies for Qualcomm's state-of-the-art SoCs. He completed Ph.D. from IIT Bombay and M.E. from the BITS Pilani, all in Electrical Engineering and has 17+ years of industrial experience. He holds several international patents, Intellectual Property white papers, Journals and presented various papers in premier IEEE conferences. He served as a program committee member of Internal Test Conference India (ITC India), Asian Test Symposium (ATS), VLSI Design (VLSID), and VDAT. He has won 2020 Journal of Electronic Testing (JETTA) - Test Technology Technical Council (TTTC) Best Paper Award. He hails from Lucknow, and cycling is his hobby.

Abstract: Since last 4 years, companies such as Google, Meta (Facebook), and Microsoft reported the manifestation of errors which, initially undetected ("silently"), had crept into their large cloud data centers. These reports mentioned that very specific instructions were intermittently incorrectly executed, propagated through the operating system, and would potentially manifest themselves as application-level errors. Many postulates have been published in the literature and since then, these systematic errors started been called as silent data errors (SDE). In this tutorial, we will begin the chapter by explaining and discussing the reports being published by the data-center companies. After collecting all essential observations, postulates, theories etc., we will try to correctly map the origin of such SDEs into various buckets such as,

1. Timing marginalities at near threshold operating voltages, with an aim to identify the extreme outlier timing paths.
2. Multi-frequency, multi-power domain designs which struggle to decide on static timing analysis (STA) sign-off corners.
3. Timing constraint generation strategies, which eventually guide the ATPG tool not to generate at-speed patterns for timing invalid paths.
4. Signal, and power integrity, localized IR drop, and voltage bump
5. CMOS reliability: bias temperature instability (BTI), hot carrier injection (HCI), interconnect RC variations etc

During tutorial, we will try to further classify whether the SDEs are test escapes and/or reliability issues? Why are they only noticed now? Is that only the case because such large server farms bring together larger numbers of CPUs than ever seen before? And what counter measures can we take against them?

TUTORIAL

Scan based Testing and Test Security Challenges: Scan and Security can Live Together

11:15am - 12:45pm

ARABICA & ROBUSTA



Jayagowri R
Professor, Dept of Electronics
and Communication
Engineering,
(PG faculty of VLSI stream)
BMS College of Engineering,
Karnataka, India

Dr. R. Jayagowri working as Professor in Department of Electronics and Communication Engineering (PG faculty of VLSI stream), BMS College of Engineering, Karnataka, India. She is the author of two books in the field of VLSI, three Indian patents and more than 70 technical research papers in highly peer reviewed conferences and journals. Her area of research interest is Low power ASIC design and Design for Testability. She is the senior member of IEEE, member IETE and life member of ISTE. She and her team won few national levels project awards in VLSI design.

Abstract: Scan based test method is considered as an efficient testing methodology for complex VLSI chips. To provide good observability and controllability at all nodes of the circuit to detect faults efficiently in the scan-based test method, it provides access to all internal nodes of the device under test. This complete and detailed access of internal nodes of the DUT in scan-based testing is desirable for testing but, this feature of scan-based testing is threat to security of the chips as it is prone to side channel attacks. So, it is mandatory to include few counter measures techniques against these side channel attacks. To achieve high fault coverage testing along with maintaining the security is challenging for engineers.

TRACK 3B

Memory Test

Session Chair : Bharath Nandakumar

Advances in Memory Design and Testing: Insights from Academia and Industry - Part I & II

PART 1: 01:45 pm - 03:15 pm

ARABICA & ROBUSTA

PART 2: 03:45pm - 05:15 pm



Sakthivel Ramachandran
Professor, Dept of Micro and Nano
Electronics,
School of Electronics Engineering,
Vellore Institute of Technology (VIT),
Vellore, India

Dr. Sakthivel Ramachandran (Senior Member, IEEE) received B.E(EEE) from Madras University, 2000 and M.E in Applied Electronics from Anna University, 2004, and the Ph.D. degree in low-power high-speed architecture development for signal processing and cryptography applications. He is currently working as a Professor with the School of Electronics Engineering, Dept of Micro and Nano Electronics, Vellore Institute of Technology (VIT), Vellore, India. He is the co-author of the book "Basic Electrical Engineering" published by Sonavarsity, 2001 and the author of the "VLSI Design" published by S. Chand in 2007. He has published more than 75 peer reviewed papers in international conferences/journals. He has delivered around 50 guest lectures/invited talk and hands on workshop in FPGA-based system design, full custom IC design, RTL to GDSII and ASIC Design, VLSI Testing and Testability (DFT), Neuromorphic computing. His research interests include low-power VLSI design, DFT, Repairable Logic Development, ASIC Design and efficient hardware

TUTORIAL

development for AI and ML. He was heading the VLSI division at VIT, Vellore, India since 2009 to 2012. During his tenure he was instrumental in setting up SoC Design Lab, FPGA/ASIC Lab, Analog IC Design Lab in association with Cadence, Altera and Texas Instruments (India) respectively.

He was a consultant/corporate trainer for Texas Instruments (Bangalore), BOSCH (Bangalore & Coimbatore), Pantech Solution (Chennai) and Wipro (Bangalore). He worked as an intern in Marvell Technologies (Bangalore). Prior to joining academia, he was the co-founder for the company Silicon to chip Technologies, Salem, India. He has mentored 30 undergraduate, 60 Post graduate students, 10 PhD students (5 completed PhDs), and 1 M.Tech (Research).

He is a part of the Fellowship chair for ITC (India) conference, recipient of AICTE ATAL fund for Silicon to chip workshop. Received the Faculty Startup Venture (FSV) Grant with the title, "GLOBE (Gateway for Lab On Basic Electronics)".



Sanjith Sleeba
Cadence India

Sanjith Sleeba is Director of Design Engineering at Qualcomm India Pvt. Ltd., where he focuses on the Memory-Test methodology and execution. After graduating in Electronics & Communication Engineering from NIT Trichy in 2001, he worked at Texas Instruments and Nvidia in different DFT roles with focus on memory testing and productization. Sanjith has co-authored multiple technical papers in various conferences and is an co-inventor of 1 US patent.



Rahul Sahu
Cadence India

Rahul Sahu has 21 years of experience in the field of SRAM Memory Design. He has expertise in SRAM low power architecture. He holds 38 US patent grants. Rahul joined Qualcomm in 2014. Prior to Qualcomm, he worked for LSI Technologies. Currently, Rahul leads SRAM memory circuit design and IO circuit design team. He holds Masters degree from IIT Delhi and Bachelors degree from BIET, Jhansi.



Subhadip Kundu
Cadence India

Subhadip is a Senior Staff Engineer at Qualcomm, specializing in silicon debug and characterization, fault modelling, and test time optimizations. Previously, he spent five years at Synopsys, developing algorithms and software for EDA-based ATPG, diagnostics solutions. Subhadip has authored over 30 journal and conference papers and has received several accolades, including the Best Paper Award at ISTFA 2021, a nomination for Best Research Paper at DAC 2013. He also holds two US patents in diagnostics and earned his PhD and master's degrees from IIT Kharagpur.

TUTORIAL

Abstract: This tutorial explores cutting-edge developments in memory design and testing, integrating perspectives from both academia and industry. It covers key topics such as SRAM architecture, sensing schemes, and design challenges at advanced technology nodes, including variability, writability, and static noise margin. The session also delves into memory fault models, test algorithms, and reliability testing strategies essential for ensuring robust memory operation.

In addition, the tutorial introduces the emerging paradigm of In-Memory Computing, where computation is performed directly within memory arrays to reduce data movement and improve energy efficiency. Finally, the tutorial presents advanced diagnosis and repair techniques, including bitmap signature analysis, redundancy-based repair, and I/O repair mechanisms, supported by a real-world case study on physical failure analysis. Attendees will gain a comprehensive understanding of modern memory systems and their evolving design, validation, and computing methodologies.

TRACK 4

Invited Tutorial Session Chair

Session Chair : Lakshmanan Balasubramanian

Scaling up DFT and Test to Bridge Today's Gaps

09:15am - 10:45am

BRAIN BOX



Rubin Parekhji
Texas Instruments

Rubin Parekhji has been with Texas Instruments (India), Bangalore, since 1996, where he has mentored and led DFT teams on various design and test technology projects across multiple groups, resulting in many DFT and test innovations, and adoption of new test technologies across TI WW. More recently, he has been in the DLP and MSP / MCU Controller organisations. He is a distinguished member of the technical staff. He has published regularly and delivered tutorials at leading conferences with several best paper awards, has mentored a large number of students, has been a visiting faculty member for multiple academic terms, has been an active TI liaison for several industry projects, has several issued patents on ideas co-authored regularly with team members, and has been on the editorial board of journals. He has a Ph.D. from Indian Institute of Technology, Bombay, India.

Abstract: The semiconductor life-cycle continues to evolve across technology, design, manufacturing and end-equipment. DFT (design for testability) and test continue to evolve to meet the quality requirements at affordable costs. As semiconductors find ubiquitous use in a variety of applications, the focus on DFT and test is shifting from being comprehensive to being ready to meet the demands of thousands of product lines and billions of chips. This talk will provide an industry perspective on what are the current gaps and how they can be bridged. Topics to be covered include (i) defect models and DFT for all-time screens, (ii) test engineering for cost reduction, and (iii) applications of AI and ML methods to manage scale. Challenges, solutions and pitfalls will also be highlighted.

ACADEMIC RESEARCH TRACK

Session Chair : Leela Krishna Thota

ART 1.1 LLMCov: A Methodology for LLM-aided DfT Coverage Improvement in Open-Source Designs

Manisha Kumari, Subharthi Roy (IIT Jodhpur, India); Masahiro Fujita (University of Tokyo, Japan); and Binod Kumar (IIT Jodhpur, India)

Abstract— Scan insertion is one of the important steps in the development of test infrastructure in modern designs. During the scan insertion step, it is very crucial to resolve all design rule violations as these violations may lead to significant drop in the test coverage. Because of nature of these violations, it becomes a tedious debugging exercise and requires lot of manual expertise. In this work, we propose AI-based solution for seeking solution for these design rule checks. The power of AI-based techniques (such as CNN-based models or LLM-based methods) is getting widely recognized in the area of EDA tools performance enhancement. Therefore, we develop a LLM-aided DRC debugging methodology to aid the scan synthesis tool that in turn assists the ATPG tool to obtain enhanced test coverage values. Experimental results indicate significant improvement in the test coverage with fixes obtained with assistance of LLM.

ART 1.2 . Optimal Test Point Insertion for Enhanced Small Delay Defect Coverage

Faraz Aatif, Prathiba Muthukrishnan and Sivanantham Sathasivam (VIT Vellore, India)

Abstract— Small Delay Defects (SDDs) often escape detection in traditional VLSI testing due to their minimal impact on circuit timing. Transition Delay Fault (TDF) Automatic Test Pattern Generation (ATPG) primarily propagates fault effects through shorter paths, which may not effectively detect SDDs. To enhance defect coverage, Test Point Insertion (TPI) can be utilized to influence the propagation of fault effect through longest path. This work proposes a novel TPI methodology that improves SDD detection by inserting test points in short paths, enabling fault propagation through longer paths where timing variations are more pronounced. Additionally, proper and strategic selection of the test points will lead to less area overhead and better delay test coverage. The proposed approach is evaluated on ITC'99 and IWLS benchmark circuits, demonstrating an improvement in delay test coverage, making it a promising technique for enhancing the reliability of VLSI testing.

ART 1.3. Assertion-Driven Formal Verification for Custom RISC-V Core

Sree Sankar E, Sakshi Thakur and Binod Kumar (IIT Jodhpur, India)

Abstract—In modern processor design, ensuring the correct functionality of each pipeline stage is crucial. Although simulation-based techniques are commonly employed to identify design flaws, they often fail to detect rare corner cases or subtle bugs that emerge only under specific conditions. This limitation poses a significant risk in safety-critical domains such as autonomous vehicles and medical devices, where failures can have severe consequences. To address these challenges, formal verification offers a mathematically rigorous and exhaustive method to validate hardware designs. This paper details a formal verification approach for a custom 5-stage pipelined RISC-V (RV32I) processor. The methodology leverages System Verilog Assertions (SVA) in conjunction with Cadence JasperGold to verify key components of the design, including instruction execution, pipeline control mechanisms, and data hazard management, ensuring reliable operation across all possible execution scenarios.

ACADEMIC RESEARCH TRACK

Session Chair : Leela Krishna Thota

ART 1.4. Enhanced Power Estimation of Digital VLSI Circuits using Machine Learning *Maniyar Mohammed Saqlain and Sakthivel Ramachandran (VIT Vellore, India)*

Abstract— Accurate power estimation remains a critical component of VLSI design, enabling valuable performance optimization. Traditional estimation techniques are not suitable for the early stages of design abstraction where speedy iteration is vital. In such a case, machine learning (ML) is a promising option with the ability to learn patterns and deliver quick, data-driven predictions. In this study, there is a comprehensive comparative evaluation of five advanced ML regressors Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Xtreme Gradient Boosting (XGBoost)—on dynamic power estimation in digital VLSI circuits. The dataset was produced through features drawn at synthesis level from designs which were processed by the Quartus Prime Design Suite to ensure usability and relevance. Model performance was analyzed through Root Mean Squared Error (RMSE) and R-squared (R²) scores in order to gauge predictive accuracy and generalizability. Among all the models, XGBoost consistently outperformed, for which the RMSE was 2.04 and the R² was 0.9888, indicating high accuracy and reliability. Compared to results earlier reported in other research, the method in this work finds an impressive estimation accuracy enhancement of about 15.2% validating the potential of ML-based techniques in speeding up and enhancing power modeling efficiency and accuracy in contemporary VLSI designing processes.

ART 1.5 Implementation of Memory Built-In Self-Test *Amol G. Patil, Suhas B Shirol and Vikas K R (KLE Technological University, India)*

Abstract— Modern integrated circuits rely more and more on embedded memories such as SRAM, DRAM, and ROM, which constitute a significant portion of SoC designs. Since these memories are designed for higher frequencies and dense data storage, they are prone to several possible faults. Traditional external ATE testing has its challenges owing to high cost and limited access to deeply embedded memories. To overcome these, this paper is proposing an efficient MBIST architecture that integrates the FSM-based controller and advanced fault simulation techniques. The solution concentrates on detecting and diagnosing the common memory faults such as stuck-at, transition, and coupling faults by implementing the MARCH X algorithm on a 16-byte memory cell. The approach promises high fault coverage with minimal overhead in terms of resources, making memory operations both reliable and fault-tolerant in advanced semiconductor designs.

PAPER SESSIONS

PAPER SESSION 1

AI in Test: Detect, Predict, Compensate

Session Chair : Prof. Jaynarayan T Tudu

1.1 Anomaly Ranking of LPDDR4 Shmoo Results using Machine Learning

Vinodh J Rakesh and Chaitanya Kumar Reddy Mallu (Infineon Technologies, India)

Abstract— Semiconductors designed for safety-critical automotive applications demand thorough verification prior to mass production. This verification process includes post-silicon characterization of semiconductor devices, such as System-on-Chip (SoC), across the boundaries of their multi-dimensional operating space. Post-silicon characterization is essential for proving interoperability and in-system performance metrics. Unlike production testing, characterization is performed on a smaller sample size, making it critical to identify anomalous SoC behavior within the characterization population, as such anomalies could indicate underlying design defects. The LPDDR4 shmoo test under consideration generates 57,600 shmoo plots, making manual analysis impractical due to volume of data, the subtlety of potential anomalies, and the time intensive nature of the task. To address this challenge, this paper proposes a novel machine-learning-based method that employs a reconstruction-based anomaly detection algorithm, specifically an autoencoder, to rank anomalous behaviour in the shmoo test results. The results of the proposed approach show that all anomalous results are ranked within the top 5% of the 57,600 shmoo test results. By focusing only on the top-ranked results, the proposed method reduces the manual effort by 95%, enabling more accurate and repeatable anomaly detection.

1.2 Machine Learning-Based High Vmin Prediction for Enhanced Pre-Silicon Reliability

Sudheer Anumala, Satya Ramoji Chikkam (Intel, India); Yu Tin Cheong, Xin Rou Chong and Mohd Azwann Zulkefle (Intel, Malaysia)

Abstract— Accurate prediction of the minimum operating voltage (V_{min}) is critical for optimizing performance, power and yield in advanced semiconductor designs. Traditional approaches rely on exhaustive simulations or post-silicon characterization, which are time-consuming and expensive. This paper proposes a supervised machine learning framework to predict high V_{min} values during the pre-silicon phase, enabling early detection of circuits prone to timing and power-related failures. The model leverages key features such as Intra-Die Variation (IDV), Leakage Current, Frequency, Temperature and Weighted Switching Activity (WSA) from at-speed ATPG patterns. We evaluate three regression models - Linear Regression, Random Forest and XGBoost using R^2 score, MAE and RMSE. Experimental results show XGBoost delivers the highest prediction accuracy. This predictive capability offers a valuable tool for identifying V_{min} -critical instances early, improving design robustness and reducing the need for costly post-silicon debug.

1.3 A Machine Learning based Built-in Self-test Method for Lifetime Frequency Drift Compensation of Precision Oscillators

Ritik Gupta (IISc Bangalore, India); Aswin R, Rubin Parekhji (Texas Instruments, India) and Gaurab Banerjee (IISc Bangalore, India)

Abstract— Embedded RC oscillators are widely used as precision reference clock generators in modern system-on-chip (SoC) designs. The lifetime frequency performance of the RC oscillators is impacted by the aging degradation of the on-chip resistance. This paper presents a machine learning (ML) based method that estimates and compensates for the lifetime frequency drift. An ML based regression model that establishes a correlation between a set of static node voltages in the oscillator and its frequency is developed. The drift is proposed to be evaluated using onchip monitoring circuits, analog to digital converter and a neural inference engine integrated in the SoC, and the calibration settings are adjusted to maintain the lifetime frequency precision. The method is validated with simulations using an oscillator circuit with a nominal frequency of 10 MHz, where the model is tested under arbitrary conditions of aging, supply voltage and ambient temperature variations, and the maximum error in estimated frequency drift is within 2000 ppm.

PAPER SESSIONS

PAPER SESSION 2

ATE Intelligence: Speed Meets Precision

Session Chair : Abhishek Chaudary

2.1 Test Time Reduction in Current Sense Amplifiers via Optimized Current Range Across Wafer Touchdowns and Gain Reordering on V93K SMT8

Hemanthkumar V, Infineon Technologies, Bengaluru, India

Abstract— In the present day, current sense amplifiers (CSA) are widely used in various applications such as power management, automotive, industrial, telecommunication etc where current monitoring is the crucial function. In this regard, testing CSA efficiently & accurately is a critical task for test engineers to ensure their performance meets the required standard. Traditional approach which keeps gain setting constant and varies current for different contracts has been noticed to consume higher test time. This paper presents an innovative approach for reducing overall test time in a wafer, by using adaptive current range across wafer touchdowns (TDs) and through gain-current reordering in a test program. The proposed approach has been verified on one of mixed signal devices on a wafer level using ADVANTEST V93K SMARTEST8(SMT8) ATE and found that test time has reduced by significant margin of about 37.8% compared to its traditional approach

2.2 Accelerating Time to Market of Semiconductor ICs using Smart ATE PCB Routing System

B Lokapriya, A Dyaneswaran, R Karthika, S Lokendran and Dhamodharan Senthilkumar (Caliber Interconnects, India) IEEE Member

Abstract— The development of Automated Test Equipment (ATE) interface PCBs is a complex and detail-intensive process. Key design tasks such as signal and power layer planning, layer stack-up optimization, tester channel assignment and high-speed signal routing are critical to ensuring the quality and functionality of the final board. Traditionally, these tasks are performed manually a practice that is not only time consuming but also prone to human error. This manual approach is increasingly unsustainable in the context of modern semiconductor advancements. With the advent of sub nanometer technology nodes, ATE interface PCBs must accommodate significantly higher pin counts and greater functional density in ICs. As complexity grows, so does the need for faster, more accurate PCB design processes that can keep pace with aggressive time-to-market demands in semiconductor product cycles. To address these challenges, we introduce a Smart ATE PCB Routing System, an automated design tool that significantly streamlines signal layer planning, layer optimization and tester channel assignment. The tool operates in two phases: Phase 1 automates signal layer planning and stack-up optimization based on user-defined strategic algorithms, while Phase 2 employs the Hungarian algorithm to perform shortest path based tester channel allocation. By adopting this automation system, design cycle time for complex ATE PCBs can be reduced by up to 90%. For instance, manual signal planning for a 1600-pin single-site device typically takes around 12 hours, whereas the proposed system completes both planning and channel assignment in under 5 minutes. This intelligent routing system supports a wide range of ATE interface board architectures, consistently delivering faster design turnaround, improved IC validation throughput, enhanced yield and reduced development costs. As a result, it fosters scalable and efficient work flows, ultimately accelerating semiconductor product delivery to the market.

2.3 Accelerating ATE Pattern Verification & Debug using Emulation Platform - Enabling First-pass Success on Silicon

Kumar Gaurav Singh, Praveen Kumar Ravichandran, Saurabh Karkun, Karthik Rajakumar, Yogeshwaran Shanmugam and Pratibha Gupta (Texas Instruments, India)

Abstract—In modern semiconductor testing, high-quality test vectors are crucial for validating the structural and functional authenticity of fabricated silicon samples. However, the conventional approach of running test vectors on simulation platforms faces significant challenges due to extremely long execution times, especially for large System-

PAPER SESSIONS

on-Chip (SoC) designs. This limitation not only impedes pattern readiness and verification but also makes debugging sequence dependencies exceptionally difficult, ultimately affecting time-to-market for new semiconductor products. This paper presents an innovative solution: The Bench Test Emulator (BTE), an emulation-based environment that bridges the gap between simulation and actual Automated Test Equipment (ATE). The proposed solution implements a simple tester model that can be deployed on both pre-silicon emulation platforms (for e.g. Palladium) and post silicon FPGA implementations. This dual-purpose approach not only accelerates test vector execution by approximately 100x but also provides a more accurate representation of the actual silicon environment, enabling first-pass success on hardware. Our methodology demonstrates significant improvements in test vector qualification time, reducing debug iterations from 40 days to just 8 days in real-world applications. The solution has been successfully implemented on multiple SoC platforms, proving its effectiveness in both functional and structural testing scenarios.

PAPER SESSION 3

Accelerated Chiplet Testing: Interconnects to JTAG

Session Chair : Rajesh Kumar Tiwari

3.1 Test Selection and Scheduling for 3D-Stacked Chiplet-based Designs

Ravikumar C P (Vinyana Tech, India) and Shiva Shankar B (IIIT Dharwad, India)

Abstract— Integrating heterogeneous blocks like processors, analog/mixed-signal circuits, and memories into a single 2D chip is complex, resource-intensive, and prone to low yields due to large die sizes. Testing such large System-on-Chip (SoC) designs is also difficult. To address these challenges, the industry now uses chiplet-based 3D integration—dividing the SoC into smaller, independently manufactured units that are later assembled. This approach improves yield and design manageability. This paper focuses on testing 3D chiplets, addressing two key optimization problems: selecting the appropriate test method for each chiplet and scheduling the tests to minimize total test time while limiting peak power. We propose a greedy heuristic based on estimated test time and power.

3.2 Effective 3D IC Interconnect Test Using Defect Distance and Bump Placement Location

Anshuman Chandra (Siemens DIS, USA); Barbara Działowska, Marta Stępniewska (Siemens DIS, Canada); Chi-Chun Yang (Siemens DIS, Taiwan); and Jonathan Gaudet (Siemens DIS, Canada)

Abstract— The power, performance and area (PPA) advantages offered by a multi-die package depend on the short, fast and low power die-to-die (D2D) interconnects that connect different chiplets in the system. Physical bumps comprising the interconnects are arranged in either rectangular or hexagonal grid patterns on the chiplet interface. These interconnects need to be properly tested for open and short defects to ensure proper connectivity between any two chiplets. Depending on the process node and the yield ramp stage, the size of the area of contamination may vary, leading to short defects beyond the immediate bump neighbors. In this paper we present an industrial flow where short defect candidate pairs are computed using (a) user-defined criteria for the area of defect for the bumps, and (b) layout information for the nets inside the interposer. Bump location information is automatically processed from a bmap file by the EDA tool and the user-defined fault model (UDFM) file is generated for ATPG to generate the patterns.

3.3 Implementation and Optimization of High-Bandwidth JTAG for Enhanced Test Time Reduction in Complex SoCs

Sandipan Sharma, Nikita Naresh (Google, India); Wilson Pradeep (Google, USA); Karthick Prabhu (Siemens DISW, Poland); and Olga Przybysz (Siemens DISW, India)

Abstract— The increasing complexity of modern System-on-Chip (SoC) designs presents a significant challenge for efficient fault testing, demanding solutions that minimize test application time while effectively utilizing Design-for-Testability (DFT) methodologies. In contrast to conventional approaches that rely on the serial application of test

PAPER SESSIONS

vectors via the IEEE Std 1687 Internal-Joint Test Action Group (IJTAG) protocol, this paper investigates the adoption of a Streaming Scan Network (SSN)- based IJTAG packet delivery methodology. This technique, termed High-Bandwidth IJTAG (HB-IJTAG), enhances testing efficiency through parallel data delivery (100X+ bandwidth gain) and enables the concurrent testing of hierarchical. This paper presents a practical case study detailing the challenges encountered during the implementation of HB-IJTAG in an industrial design environment and the strategies developed to mitigate these issues. Furthermore, architectural enhancements to the SSN based IJTAG packet delivery are proposed to achieve additional reductions in test application time. The efficacy of this methodology is demonstrated through evaluation results obtained from the industrial design highlighting a significant reduction in test time by 80-90% on certain type of tests compared to traditional testing approaches.

PAPER SESSION 4

AI in Test: Detect, Predict, Compensate

Session Chair : Nikita Naresh

4.1 From Structural Test Escapes to Silent Data Errors: A preliminary analysis

Paolo Bernardi, Francesco Angione (Politecnico di Torino, Italy); and Arani Sinha (Intel, USA)

Abstract— In recent years, the observed Failure-in-Time (FIT) rates of semiconductor devices in data center fleets by cloud service providers have drastically increased, pinpointing that defective devices are escaping during manufacturing tests. Although semiconductor manufacturers minimize FIT rates of their devices, the exponential deployment at scale, combined with growing design complexity and increasing transistor density, results in a non-negligible number of defective devices in server fleets. High-end processors undergo different manufacturing test phases to ensure low defect-part-per-million (DPPM) of shipped defective devices. However, it is always difficult, if not impossible, to cover the full spectrum of all possible defects in a device. This paper proposes a grading methodology for manufacturing test escapes of a permanent nature, such as Stuck-At faults, Transition Delay faults, and Small Delay Faults, that would be very likely to produce Silent Data Errors (SDEs). The proposed methodology combines several structural measurements that describe a fault's likelihood to create a silent data error. If the faults that are at high risk of being SDE can be identified early in the product lifecycle, design decisions can be made to prevent these faults from actually creating silent data errors when the device is deployed. Experimental results are carried out on arithmetic modules used frequently in High Performance Computing.

4.2 Lite-CMOS Transition Fault Model (CTFM) for Legacy Design(s)

Yi Sun (NXP Semiconductors, USA), Qiong Wang (NXP Semiconductors, China), Saidapet Ramesh (NXP Semiconductors, USA) and Cunyu Shi (NXP Semiconductors, China)

Abstract— Advanced chips, built on innovative process nodes, are more susceptible to defects during manufacturing, making early detection critical. This necessity has driven the demand for extensive structural scan testing to screen for defects in the digital logic of System-on-Chips. The widely adopted Stuck-At (SA) and Transition-Delay (TD) scan ATPG fault models are not sufficient for screening out the defective parts and achieving low DPPM automotive parts. Targeted pattern generation for the SA and TD fault models is done only using port-level faults. Hence, the cell internal defects are not always visible via SA and TD patterns. The more recently developed Cell-Aware (CA) fault model has been successfully proven to be able to screen out the cell-internal defects. However, this approach incurs significant overhead, including CA-UDFM (user-defined fault model) generation. In this work, we present CTFM (CMOS Transition Fault Model), which uses custom UDFM files that target cell internal faults at the transistor level with less overhead compared with CA patterns. This is crucial for legacy designs, where fewer resources are typically allocated, especially compared to New Product Introduction (NPI) designs. A custom in-house diagnosis flow using our UDFM to improve scan diagnosis quality is also presented in this work.

PAPER SESSIONS

4.3 Assessing Long-Term Reliability through Accelerated Aging Stress Tests on a On-chip Quality and Reliability Platform : A Band Gap Reference case study

Shiv Shankar Gupta, Javed GS and Rahul Sharma (Intel, India)

Abstract— The accelerated aging stress test is essential for assessing the reliability and quality of band gap reference (BGR) circuits, which are crucial for maintaining consistent voltage references in ICs. In this study, we present accelerated aging analysis of a classic BGR design as an case study. A BGR circuit, validated on a Quality and Reliability On-Chip (QROC) platform, demonstrated temperature coefficient of 0.12 mV/oC across 25oC to 100oC, ensuring minimal drift with temperature fluctuations. The output reference voltage (Vref) 646mV was measured over a supply voltage range of 0.8V to 1.4V, with a 7.5ppm/V. The circuit's robustness was further confirmed by observing only a 1% change in Vref after a million-second stress test at varying temperatures and a stress voltage 1.4V. This result, showing less than 6.5% deviation from post-layout simulations, underscores the BGR's resilience and precision, highlighting the importance of such stress testing in predicting IC performance and longevity

PAPER SESSION 5

ATE Intelligence: Speed Meets Precision

Session Chair : Rahul Anand

5.1 VTEST: FPGA-Based SoC Validation Framework

Srikanth Neelam, Sreekanth Arati (VConnecTech Systems Pvt. Ltd, India); and Amalin Prince A (BITS Pilani, India)

Abstract— Post-silicon validation of System-on-Chip (SoC) designs is traditionally dependent on a wide array of external hardware tools, software utilities, and host-controlled workflows to evaluate power, clocking, functional logic, and debug interfaces. These fragmented setups often involve separate controllers for power sequencing, JTAG probing, clock programming, and firmware loading, which increases validation time, complexity, and the likelihood of interface mismatches. Moreover, the lack of runtime reconfigurability and signal level observability limits early-stage silicon bring-up, especially when multiple voltage rails and interface standards are involved. To address these challenges, an FPGA-based validation framework is proposed that unifies multiple validation paths through a compact and reconfigurable setup. Power rail sequencing and voltage setup is achieved by configuring a single I2C slave PMIC to drive multiple outputs, enabling full power-up validation across SoC domains. On-chip temperature sensing and current monitoring are also integrated for health checks. Multiple JTAG and boundary scan interfaces present on the SoC are accessed through a common FPGA-controlled JTAG interface, enabling ATPG, BIST, and debug module validation without requiring separate hardware connections. Clock sources across the SoC are configured dynamically using two-wire I2C or SPI lines to programmable oscillators, allowing flexible frequency programming without external tools. Reset lines are controlled through programmable logic, supporting glitch injection, sequencing, and controlled de-assertion for robust startup validation. Firmware loading is accomplished over Ethernet, while debug visibility is provided through integrated logic analyzers and UART-based logging—eliminating the need for software driven utilities. By consolidating all validation stages—from power integrity to protocol and debug analysis—into a single FPGA-controlled environment, the framework enables high observability, reduces hardware dependency, and supports full automation. Leakage and operational current profiling, trace capture, and firmware execution tracking are seamlessly handled through minimal signal routing. This approach significantly reduces the need for protocol converters and debug utilities, offering a scalable and efficient solution for validating modern SoC designs in both engineering and production environments.

5.2 Novel approach for implementation and validation of Scan Dump debug methodology

Savan Bhatelia and Mahesh Bendre (Intel, India)

Abstract— Scan Dump is useful debug technique used for debugging functional failures during field operation or in the lab. Typically, combination of in-house HW and SW based methodology is used to implement Scan Dump, but in-house methodology poses significant challenges for complete verification of scheme and scalability across different

PAPER SESSIONS

SoCs. This paper describes a novel and scalable approach to implement and verify Scan Dump operation. Approach involves inserting H/W components required for Scan Dump operation along with regular structural DFT components like Scan Streaming Network (SSN), Embedded Deterministic Test (EDT) and On Chip Clock Controllers (OCC) at physical PNR block level. Custom H/W connections are made to merge Scan Dump operation path into structural test IJTAG network. Scheme allows to verify Scan Dump operation completely at partition level without depending on SoC level infrastructure. All required validation collaterals can be generated at partition level and retargeted at SoC level.

5.3 Novel Post Silicon Validation Strategies for the New Era of UCle Chiplets.

Bhuvana Chandar, Anirudha Rao, Santhosh Kumar S, Meghana Bukkapatnam, Deepak Baliga (Intel, India); and Chad E Beach (Intel, USA)

Abstract— The Universal Chiplet Interconnect Express (UCle) standard marks a new era in SoC design by enabling the seamless integration of heterogeneous chiplets from multiple suppliers and process nodes into a single package. However, post-silicon validation of UCle IP poses challenges due to the inpackage high-speed I/O links and lack of BKM's for this new technology. Our team validated Intel's first UCle-compliant silicon, and in this paper, we outline the challenges and solutions across various post-silicon validation functions. Our learnings will help provide robust die-to-die link testing on the many upcoming products that have adopted our UCle IP.

PAPER SESSION 6

Accelerated Chiplet Testing: Interconnects to IJTAG

Session Chair : Kiran Rajmohan

6.1 An Integrated Environment for Simulation of Mixed Signal Tests using Open-Source Tools

Gursimran Singh, Irawati Thete, Pammy Thakur and Rajesh Jabade (Texas Instruments, Bangalore)

Abstract— As mixed-signal integrated circuits become more complex, testing them takes more effort, cost and time. This makes it imperative to simulate mixed-signal tests before silicon arrival however this has never been practical except for the smallest of test-cases. This results in errors in post-silicon test procedures that present a laborious and time-intensive task. This paper shows how Verilog Procedural Interface in conjunction with Python can be leveraged to create an integrated environment that enables simulations of mixed-signal ATE test procedures presilicon. Examples of how the environment plugs into the existing design flow without any disruptions and some use cases are also presented.

6.2 A Smart Framework for Cost-Effective Characterization of Touch-Sensing Controller

Paul Ngoy (Infineon Technologies, USA); Abhishek Ganesh Poojary, Jagadish Raju K and Keerthan Rai (Infineon Technologies, India)

Abstract— Conventional methods for Touch Sensing Controller (TSC) characterization are prone to inaccuracies with modern touch panels and become expensive due to prolonged test time. To address these challenges, a cost-effective CPU-based solution is proposed for real-time system-level performance analysis. The approach mitigates serial communication bottlenecks and reduces dependence on large on-chip memory. The solution was validated on a TSC with 100 16-bit Capacitance-to-Digital Converters (CDC), 8 K samples per CDC and 96 KB of on-chip memory, achieving 94% improvement in memory utilization and 84% reduction in test cycle time. By closely aligning characterization setups with real-world application conditions and optimizing the data acquisition, the proposed method enhances timing accuracy and, ensures robust analysis.

PAPER SESSIONS

6.3. Automated BIST Technique for Interconnect Characterization and Reliability Enhancement in Flexible Electronic Circuits

Harshad Singh, Kirti Srivastava and Oppili Prasad L (IIT-BHU, India)

Abstract— Electronic circuits used in Flexible Electronics are often prone to reliability issues due to mechanical stresses exerted on the substrate material during operation. Such flexible electronic circuits often include meandered interconnect wirings routed on flexible and stretchable substrate materials such as elastomers, polymers, etc. Mechanical stresses exerted on the substrate material can cause the interconnect wires to deform, thereby significantly affecting the electrical properties of the interconnect routes such as resistance, parasitic capacitance, inductance, etc. Extreme levels of mechanical stresses in the system can lead to open circuit faults in the interconnect routes. Hence, it becomes imperative to monitor the health of interconnects in such flexible electronic circuits, so that appropriate automated repair actions can be taken, whenever there are any reliability issues in the circuits. In this work, we propose an automated Built-in Self Test (BIST) technique to monitor the variations in resistances of interconnects. The BIST technique also involves automated steps to take appropriate measures to active a redundant subcircuit present in the system, whenever faults are diagnosed in the interconnects, such that the circuits are kept functional.

PAPER SESSION 7

Smart SoC Testing: Access, Memory, Reliability Session Chair : Sudhir Raut

7.1. Exploring Tessent Two Pin Serial Port for MBIST, LBIST and ATPG for a low pin count SoC

Ratheesh Thekke Veetil, Soumyajyoti Paul, Prakash Kumar, Anvesh Gadi (Analog Device, India); Ajay Purushotham and Vidya Neerkundar (Siemens EDA, India)

Abstract— In modern chip design, optimizing pin count is critical to meeting size, cost, and performance constraints, particularly in low-pin-count, low-power SoC (System on a Chip) designs. This paper presents an efficient solution to the challenge of enabling multiple test functionalities, such as MBIST (Memory Built-In Self-Test), LBIST (Logic Built-In Self-Test), and ATPG (Automated Test Pattern Generation), through a minimal two pin interface. We propose the use of TPSP (Two Pin Serial Port), a tool-supported interface that facilitates the generation of low power test patterns using commercially available EDA (Electronic Design Automation) tool, ensuring seamless integration into our 5nm low-power SoC design. This solution optimizes pin usage, enhances power efficiency, and maintains robust test coverage. Silicon results validate the approach, showing excellent correlation between pre-silicon and post-silicon power predictions, with all test patterns passing on day one.

7.2. Enhanced RTL Test Point Insertion Flow with Improved Functional Block Testability

Nikita Naresh, Naushad Ali (Google, India); Wilson Pradeep (Google, USA); Oussama Laouamri (Siemens USA); Surya Phanindra Chatla, Karthick Prabhu B (Siemens India); Nilanjan Mukherjee (Siemens USA)

Abstract— This paper introduces a novel methodology to extend Register Transfer Level (RTL) test point insertion (TPI) capabilities to functional blocks (FB) encompassing operators such as comparators, arithmetic and logical reduction operators, and shifters. The proposed methodology leverages a function-call-based technique for precise back-annotation of functional block expressions and the insertion of test point logic at targeted nodes. By effectively isolating complex regions within RTL designs and enabling the placement of control points and observe points, this approach achieves higher granularity in improving test coverage while preserving design optimization during synthesis. Additionally, it addresses challenges related to expression and logic sharing, ensuring correctness when RTL contexts overlap or diverge, with minimal impact on design area and performance, post-testpoint insertion. Furthermore, the methodology provides actionable feedback to RTL designers to identify locations unsuitable for test point insertion, thereby refining testability at RTL. Experimental results demonstrate the advantages of enhanced TPI to functional blocks which include reduced pattern count and minimal area overhead in comparison to gate level (GL) TPI.

PAPER SESSIONS

7.3 Enhancing Memory Reliability: Integrating Error Correction Codes with Memory Built-In Self-Test for Effective Validation

Luc Romain (Siemens, Canada); Paul-Patrick Nordmann (Siemens, Germany); Albert Au (Siemens, Canada); Wei Zou, Harshitha Kodali, Martin Keim and Lori Schramm (Siemens, USA)

Abstract— Error correction codes (ECC) are techniques for detecting and correcting errors. The adoption of ECC is increasingly prevalent, driven by its ability to manage random defects in emerging memory technologies and to mitigate soft errors. When testing memories with ECC, the memory builtin self-test (BIST) logic must account for ECC's presence to accurately report the status during both manufacturing and insystem test. This paper details the innovations to memory BIST logic and the testing sequences to validate memories integrated with ECC, including those with redundancy.

PAPER SESSION 8

Secure and Efficient SoC Test Infrastructure

Session Chair : Rajesh Khurana

8.1 Hybrid Bscan and Self-Test IOs to Reduce IO Test Time

Mathangi Raghuraman, Sujith Thomas, Sharmila Jayakumar and Ramakrishnan Subramanian (Sandisk, India)

Abstract— As the SoC market becomes increasingly dynamic, cost optimization is crucial for success. Reducing test costs, especially for IOs, plays a significant role. While parametric and boundary scan methods are both used for IO testing, each has its drawbacks. This paper presents a novel Hybrid Bscan methodology for testing IOs, combining the fast and cost-effective boundary scan with the fault coverage of direct parametric testing via electrical probing. Silicon results on 7nm IO testing show a 5.4x reduction in test time for common IO specifications. As a further enhancement, we introduce 'Bscan with self-test IOs' method, which uses the comparator in high-speed IOs as an ADC, eliminating the need for electrical probing while maintaining fault coverage of parametric testing. Silicon measurement of 7nm IO on-die termination resistance using this approach yields a further 6x reduction in test time.

8.2 Concurrent ATE testing of MSIP PHYs in SOC via Scan Streaming Fabric

Kriti Sundar Das, Paras Jain (Google LLC, India); and Harini Kulkarni (AMD, India)

Abstract— Modern System-on-Chips (SOCs) are increasingly complex, integrating a diverse array of digital, analog, and mixed-signal IP blocks, including high-speed PHYs like PCIe, USB, UFS, and HBM. Ensuring the quality and reliability of these on Silicon requires numerous test methodologies, spanning Slow-ATPG, At-speed ATPG, MBIST, memory repair, and dedicated tests for analog components like PLLs, FLLs, and the various PHY interfaces. While test patterns for the core digital logic, such as ATPG and MBIST, often contribute a smaller portion to the overall test duration, the testing of complex analog blocks and high-speed mixed-signal PHYs typically constitutes a significant fraction of the total ATE test time. In this Paper we present 2 methods of Testing of the PHYs concurrently on ATE : Scheduling and Scan-Streaming Fabric and thereby saving the Test-time by 50-70% for PHY Analog Tests. We also present a seamless flow of pattern porting from PHY IP level to SOC level via JTAG 1687 IEEE protocol and a bridge module for efficient verification and man-hours optimization .

8.3. FuseLock: Dynamic Taint-Based Scan Chain Protection with Fuse-Guarded Counters

Niranjana Ithal, Gautham Varma Kanumuru, Lasya Chidanand Hegde, Tanish Shet and Sudeendra Kumar K (PES University, India)

Abstract— Present-day Integrated circuits require extensive testing to ensure functionality and efficiency, often outsourced to semiconductor assembly foundries and test (OSAT) facilities.

PAPER SESSIONS

However, this exposure of IP introduces consequential security risks, such as reverse engineering and malicious pattern insertion leading to IP theft. A wide range of solutions have been explored to mitigate these risks, attempting to efficiently streamline the process of testing IC's without having to expose it to third party untrusted vendors. Unfortunately, most of these solutions are accompanied with high test time complexity, test coverage loss, or inconvenient test procedures. To address this, we propose a holistic scan chain security architecture that combines per pattern authentication and selective scan response encryption, tailored for secure testing in untrusted environments. Our work also enables functional testing at OSAT without compromising on test security, avoiding the need to ship the packaged ICs to the design house for activation or for fusing security key/seed. We also use the counter-based mechanism that detects tampering and limits the testing attempts. Our work seamlessly integrates with existing test flow while also providing strong resistance against malicious pattern insertion and threats posed by untrusted third party OSAT, with minimal impact on area and performance.

PAPER SESSION 9

Test Cost Optimization: Scan and LBIST

Session Chair : Anirudh Kadiyala

9.1 A Left Shifted Approach for Low Shift Power ATPG Decompression

Hillol Maity, Hanumant Tuntuni, Sreenu Kakunuri (Synopsys, India); Peter Wohl (Synopsys, USA); and Parthajit Bhattacharya (Synopsys, India)

Abstract— Controlling power consumption during VLSI testing is critical for modern nanoscale technologies, particularly managing shift power within test compression frameworks. Conventional ATPG methods struggle to meet strict shift power budgets effectively when compression logic limits direct scan chain access. This paper introduces a novel power-aware ATPG technique centered on horizontal (scan-cells at same shift position) transition control, integrated seamlessly with decompressor logic. We propose a hardware-assisted mechanism mimicking adjacency fill to enable deterministic shift-hold capabilities via dedicated MUX logic controlled by ATPG. Furthermore, we detail a "Shift Left Strategy" employing ATPG guidance to proactively inform the test generator about established hold window constraints, preventing care-bit conflicts and significantly improving fault merging under tight power budgets. Experimental results on production designs demonstrate that, compared to a reference low-power technique, our approach achieves substantial reductions in test patterns (avg. 2.02x), test cycles (avg. 1.6x), and ATPG runtime (avg. 1.65x), alongside an average test coverage improvement of 0.79%, validating its effectiveness for stringent shift power targets (down to 25%).

9.2 Optimized Grouping for Packetized Scan Solutions

Bhavika Ranjeet Kumar, Aalhad Deshpande, Rama Sireesha Arisetti and Maheedhar Jalsutram (Google, India)

Abstract— The increasing complexity and core count of modern System-on-Chips (SoCs) necessitates innovative testing methodologies. Traditional methods, constrained by limited pin count, struggle to significantly reduce test time and data volume. Packetized scan networks offer a promising solution by enabling efficient, high-speed, concurrent testing of multiple cores through packetized data streaming at higher delivery rate. However, effectively leveraging packetized scan solutions for optimal test efficiency requires careful consideration of core grouping strategies. Randomized grouping of cores can lead to inefficient testing and potential IOP overload and IR drop issues. The distribution of scan volume across cores is highly non-uniform, and combining cores with significant differences in scan volume leads to an increase in wasted bits. This paper proposes a novel, constraint-aware core grouping mechanism for concurrent testing that maximizes test efficiency while adhering to design constraints. Our methodology considers factors such as power clamp limits, thermal distribution, and test data overhead (wasted bits). By strategically clustering cores with compatible test requirements, we minimize test time and facilitate higher internal core frequencies during testing. This approach ensures efficient resource utilization while adhering to power and thermal constraints, ultimately enhancing the testability of complex SoCs. This methodology has been used to create core groupings for a real-world SoC, and the results have been validated on silicon.

PAPER SESSIONS

9.3 Tester time savings for LBIST diagnostics using enhanced version of On-chip MISR signature comparison with self-test and bypass

Jaidev Shenoy, Kelly Ockunzzi (Marvell Technology Inc.) and Virendra Singh (IIT Bombay, India)

Abstract— With the shrinking time-to-market schedule, it is very important to be able to detect manufacturing defects and rectify the fabrication processes faster in order to have better quality products during mass production. Compression techniques have been adopted to shorten the scan chain lengths and in turn reduce the test time but, the data loss during signature generation makes the diagnosis harder and the difficulty increases with high data compaction techniques like Multiple- Input-Signature-Registers(MISRs) which are predominantly used in Logic Built-In-Self-Test(LBIST). These type of compactors are plagued by poor diagnostics, since it is very difficult to identify the locations of error-capturing flops per LBIST pattern basis which is the pre-requisite for locating the exact fault site. Though onchip MISR signature comparison methods with few extra gates have been proposed before to improve MISR diagnosability, the method of identifying faults in the added circuitry itself have not been analyzed. In this paper, we have introduced a self-test process to tackle this issue, along with a bypass mechanism to prevent yield loss if there are manufacturing defects in the extra logic. We have also introduced the on-chip compare method in addition to traditional LBIST in this paper and shown significant tester time savings during LBIST diagnostics.

PAPER SESSION 10

Secure Resilient Fabrics: Designing for Reliability and Unclonability Session Chair : Prof. Binod Kumar

10.1 EFH - An Efficient Fault-Tolerant Routing Methodology for 2D Mesh NoCs

Biswajit Bhowmik, Girish K K, Akot Jayesh Raju and Rohit Chakraborty (NIT Karnataka, India)

Abstract— The increasing complexity of modern System-on-Chip (SoC) designs demands reliable and scalable communication frameworks. Network-on-Chip (NoC) architectures, particularly the 2D Mesh topology, have gained prominence due to their structured layout and scalability, facilitating efficient data routing among interconnected functional blocks. However, the 2D Mesh topology remains highly vulnerable to static and dynamic faults, which disrupt network performance and increase congestion. Existing fault-tolerant routing algorithms struggle with handling complex fault patterns, such as concave and irregular fault regions, leading to increased latency and packet loss. This paper introduces the Entrance, First, Hole (EFH) routing methodology, which employs a dynamic node-labeling strategy to classify nodes into entrance, first, and hole nodes. This classification enables the network to bypass faults, maintain functional nodes, and optimize routing paths without introducing significant overhead. Experimental results using the Noxim NoC simulator demonstrate that EFH significantly enhances network throughput, reduces latency, and improves overall fault resilience compared to existing approaches.

10.2 . Design and Analysis of Complementary Cascode Current Mirror based Physically Unclonable Function using 65nm Predictive Technology Model

Shreeji Shah and Rajesh Thakker (Gujarat Technological University, India)

Abstract—Physically unclonable functions (PUF) are important primitive for hardware-based security. An improved PUF cell based on Complementary Cascode Current Mirror with Muller-C element is proposed. Analysis for different PUF parameters is done including native bit stability, bit alias, uniqueness, and uniformity against varied temperature and voltage range. The design and simulation are carried out in LTSpice using 65nm Predictive Technology Model. The proposed design stands out in terms of unstable bits and is good as to available designs in terms of uniqueness and uniformity.

POSTER SESSION

1. A Hybrid Strategy to Design Efficient Extest Compression at RTL

Subrata Roy, Saravanan Balakumar, Syed Ali Om, Gopi Bobba, Subrata Roy and Vaibhav Srivastava (Nvidia Graphics Private Limited India), Karthick Prabhu(Siemens DISW, Bengaluru, Karnataka), Oussama Laouamri and Nilanjan Mukherjee (Siemens DISW, Wilsonville)

Abstract— To create accurate EXTEST compression logic for cluster-based designs, it is essential to determine the wrapper count at the RTL (Register Transfer Level). This paper introduces a novel method to calculate both shared and dedicated wrapper counts at the RTL and to design efficient compression logic based on these estimates. We describe blocking techniques and wrapper analysis at the RTL, which yielded comparable results by utilizing industry standard wrapper analysis feature and Nvidia internal CAD (Computer Aided Design) flow enhancements. This method provides precise wrapper cell count estimations, enabling accurate EDT sizing for EXTEST designs. By eliminating the need for iterative feedback, it enhances design quality, reduces area overhead, and decreases test time and turnaround time in the DFT (Design for Test) implementation and verification cycle. This approach simplifies the chip design process, leading to better design quality at earlier stages and increased efficiency overall. Runtime for computing the wrapper flops between RTL and Gate level will be demonstrated along with how many dedicated and shared wrapper are identified.

2. Boosting Scan Test Efficiency: Optimized ATPG QoR for Superior Test Coverage & Quality

Darshan V, Karthikeyan Soundararajan , Praveen Palani and Arul Karthick Kumar (Qualcomm India Pvt Ltd)

Abstract— Modern System-on-Chip (SoC) designs are engineered to handle increasingly complex tasks while delivering enhanced performance within a small form-factor and maintaining low power requirements. As the complexity of cutting-edge SoCs continues to scale drastically, achieving high quality test coverage, optimal test pattern sets, and low average unit cost becomes imperative. This paper addresses the challenges associated with achieving optimal Automatic Test Pattern Generation (ATPG) Quality of Results (QoR) in such advanced SoC designs. The proposed solutions address challenges in achieving high test coverage at reduced test cost.

3. Hierarchical Test Using Running MISR Signatures

Dale Meehl, Brion Keller, Krishna Chakravadhanula and Pradeep Nagaraj (Cadence Design Systems San Jose, CA, USA)

Abstract— Most large chip designs today utilize a Core-based testing methodology. Most Core-based test methodologies utilize DFT logic from chip testing within each Core and use boundary isolation logic to make testing of the Cores independent from each other. One of the older chip testing approaches was to use MISRs - sometimes for Logic BIST and sometimes for test compression of ATPG patterns. Except for LBIST use of MISRs for Core testing is not common. In this paper we show a DFT approach in which MISRs are used as the test compression mechanism within Cores utilizing an LBIST-like running signature so that the signatures do not have to be checked very frequently.

POSTER SESSION

4. Enhancements in CA-UDFM Models to Optimize the Development Flow of Custom Macros/IP and Standard Cell Libraries

Ravi J N and Pramod Gayakwad (NXP, India), Andreas Glowatz (Siemens, Germany)

Abstract— Cell-aware (CA) testing has gained prominence in ensuring the reliability and robustness of integrated circuits (ICs), particularly for achieving low Defective Parts Per Million (DPPM) targets in automotive and other safety-critical applications. As the complexity of semiconductor designs increases, traditional fault models such as stuck-at and transition faults are proving inadequate. This paper presents advancements in User Defined Fault Models (UDFM) tailored to address latent defects within custom macros, IPs, and standard cell libraries. By incorporating novel methodologies such as visualization of Layout Parasitic Extracted (LPE) netlists, critical region identification, and Cell Neighborhood Bridge (CNB) analysis, defect coverage is enhanced, and design-for-testability (DFT) is significantly improved. The proposed methodologies have demonstrated substantial reductions in design cycle time while ensuring better silicon reliability and manufacturability.

5. Leveraging TMU sensors for Software-driven Thermal Mapping and Visualization in SoCs

Akhilesh Shenoy and Shradha Todi (Samsung Semiconductor India R&D)

Abstract— Efficient thermal management is critical for ensuring the functional reliability, performance stability and lifetime optimization of modern System-On-Chip (SoC) designs. This paper presents a novel software-driven approach for realtime visualization of temperature variations across different regions of an SoC using the embedded Thermal Management Unit (TMU) sensors – eliminating the need of any external thermal imaging hardware, such as infrared (IR) cameras. The approach proposed in this paper, utilizes a Python-based workflow to transform raw sensor outputs into spatially accurate and interactive heatmaps, which are scalable, architecture-agnostic and compatible with any platform featuring a TMU-compliant driver. This enables designers, validation engineers, and test teams to non-invasively monitor thermal profiles, detect potential hotspots, correlate thermal behavior with workload activity, and enhance heat dissipation mechanisms during post-silicon debug. By focusing solely on the data collected from TMUs, this approach provides a cost-effective and efficient alternative to traditional thermal imaging. The visualization enhances the accessibility and usability of thermal data, supporting more thermal-aware design practices, contributing to a longer device lifespan and improving overall system efficiency.

6. Inline Shmoo Classifier using Deep Learning Algorithm

Rohit Sirwani, Harish K Mallya and Tavishi Dutt (Intel Corporation India)

Abstract— There is a need to reduce the debug time of post silicon failures without compromising on quality. Inline shmoo Classifier is one such step in that direction. It is an end-to-end pipeline without any human interference to capture the shmoo during the test instance run and identify its behavior with the help of state of art Deep Neural Networks. This methodology will aid the debug engineer in validating the failure parts and further reduce the debug time by adjusting the test program flow based on the prediction. This will also help to reduce product qualification to socket elimination timeline with faster debug outputs. In this paper, we detail the predictive model used, its performance during the training

POSTER SESSION

phase, how it has been integrated into the testing workflow, the outcomes achieved, and the prospects for future enhancements. The successful implementation of this system demonstrates the potential for incorporating cutting-edge Deep Neural Networks models into testing procedures for various applications, and it establishes a framework for such technological integrations.

7. Programmable Strobe Generator for High Bandwidth Memory (HBM) Input/Output Built-In-Self-Test (IOBIST) Controller

Sathya Kulkarni (NVIDIA Corporation, United States), Snehitha Gunda and Amanulla Khan(NVIDIA Graphics Pvt. Ltd, India) , Pradeep Vanama om and Aseem Jain (NVIDIA Corporation, United States)

Abstract— HBM is a high throughput, low latency DRAM used in datacenter GPUs (Graphics Processing Units). Due to large number of IOs and complex packaging technology, the probability of assembly defects on the connections between HBM and GPU increases. These defects are identified using an IOBIST Engine. As the HBM standard evolves, its features like burst length, number of preamble and postamble cycles on write and read strobes, etc., keep changing. This required microarchitecture work, RTL design and verification of IOBIST Engine for every generation of HBM, which is proven inefficient. This paper summarizes a scalable programmable strobe generator microarchitecture which alleviates the problem.

8. Effective In-field Test with Autonomous In-System Test Controller for Safety Critical Applications

Kranthi Kandula, Harsha Balan M, Ramalingam Kolisetti, Shamitha Rao and Jay Agrawal (Synopsys India Pvt. Ltd, India)

Abstract— Automotive and industrial applications are safety critical and must comply to safety regulations such as ISO 26262 (automotive) and IEC 61508 (industrial) respectively. Highquality silicon must have zero defective parts per million (DPPM) to ensure no faulty chips are shipped to customers. Periodic In-field testing is required to ensure safety and reliability. To achieve the highest ASIL levels, automotive SoC makers constantly face new challenges related to manufacturing testing and support for in-field tests. In this paper, we present a novel architecture of Autonomous In-system Test (AIT) with Xtolerant Logic BIST for an effective self-driven in-system test with power-on-self-test, in-system test and manufacturing test that also meets the quality and reliability standards defined by ISO 26262, while ensuring effective test scheduling and test time reduction.

9. AI for Test Optimisation to enable best out-of-the box Test QoR

Pooja Vishwanath, Greeshma Jayakumar(Synopsys India), Sruthi Nanduru (Synopsys, United States), Jose Angel (Synopsys, Mexico), Zhiwei Liao and Sri Ganta(Synopsys, United States)

Abstract— The increasing functionality in today's chips demands high quality, often involving multiple design iterations to identify optimal test parameters. Finding an optimal test configuration that is customized for each design core is often time-consuming and resource-intensive. This paper proposes an approach leveraging AI to optimize DFT planning as well as ATPG pattern generation, resulting in fewer

POSTER SESSION

production patterns and therefore, test costs. This capability allows DFT architects to anticipate optimal test parameter ranges way ahead in the design cycle, significantly reducing the need for extensive trial-and-error experimentation. This approach has been shown to achieve a consistent test cost reduction across all application segments, with a typical 15% to 20% pattern count reduction.

10. Enhanced Power aware DFT Methodologies for Complex Multi-Power Domain SoCs

Tathagata (Google, India), Wilson Pradeep(Google, United States)

Abstract— With the introduction of ultra-low power SoCs, designers are creating more functional power domains in Unified Power Format (UPF) to meet the stringent power metrics. This poses a challenge to DFT engineers as many unique power crossings may get introduced post DFT insertion and connections. DFT engineers have to make sure that these DFT paths are complied with IEEE 1801 correctness standards (UPF). Existing EDA solutions and methodologies cater to some of these challenges with correct by construct implementation, but still many corner cases are seen in real designs as described here, which requires additional efforts. This paper describes the limitations of existing EDA solutions, and derives new automations to correct them. This paper also propose a novel methodology to auto create UPF, and fix the missing functional strategy(s) in the input UPF without introducing redundancy and QoR degradation, thereby reducing CLP violations and improving execution turn-around time

11. Realization of TMDS Protocol on FPGA for HighSpeed Data Transmission Applications

Ganesh Racha, Karthik Gupta Telukunta, Yedukondalu Kamatham, Murtaza Ahmed Farooqui, Srinivasa Rao Perumalla and Nagireddy Manaswini (Department of Electronics and Communication Engineering CVR College of Engineering Hyderabad, India)

Abstract— In 4G/5G and beyond communication applications, there is a thrust for high-speed data transmission using analog/digital interfaces and audio/video display technologies. Transition Minimized Differential Signaling (TMDS) is a technology used in HDMI, DVI, and Display Port interfaces for providing efficient, high-bandwidth communications while minimizing electromagnetic interferences. This work proposes an algorithm for encoder/decoder logic for flexibility, parallel processing capabilities, and scalability to use in high-speed communication systems. This design process uses Verilog HDL for hardware description, simulation, and synthesis for targeting AMD Spartan-7 FPGA and Vivado platform. The results indicate that the design effectively supports full HD 1080p video transmission.

12. Advanced SIR Testing for Trapped Solder Paste Flux Residue

Aurkie Ray (Macdermid Alpha Electronics Solutions, India)

Abstract— One of the biggest challenges to producing electrochemically reliable electronics is the trapping of flux residues under Bottom Termination Components (BTC), such as QFN's, MLF's and some LGA's. BTC use has become extensive in electronic assembly due to low cost and small size with improved thermal and electrical performance, making these components very attractive for many applications. Electrochemical reliability of the BTC assemblies has been addressed in many publications for both cleaned and unclean electronic assemblies in the recent past. Many no-clean solder pastes that are very

POSTER SESSION

reliable in standard SMT reflow assembly processes with non-BTC configurations exhibit electrochemical failures under BTC's, especially with a larger ground pads/heat sink design. Solder paste flux activators and solvents are unable to evaporate during reflow; they remain behind in the flux residue and allow electrochemical failures, such as low Surface Insulation Resistance (SIR) and Electro Chemical Migration (ECM) shorting. A second reliability challenge is the ever decreasing spacing between components, leads and pitch reduction overall. Mobile and communication devices present the worst combination of these reliability challenges because of their high board density. There is a need to develop solder pastes that not only allow printing small stencil apertures but also provide better electrochemical reliability when flux residue is trapped under low-standoff components with closely spaced IO's. To meet this need, an SIR/ECM test methodology has been developed using 200µm spacing 1 mm trace comb patterns covered with thin, light glass slips that float on top of the solder paste during reflow that prevent the volatilization of most of the paste activators and solvents and keep them within the flux residue. These test vehicles are then exposed to 85°C and 85% RH at 10V. Most commercially available solder pastes showed some degree of ECM within hours of exposure. The development of this test method has allowed better differentiation between the reliability of various solder paste and facilitated the development of pastes that are much more reliable under BTC's and take a significantly longer time to fail.

13. Reimagining ATE Efficiency: A TRIZ-Driven Methodology for ATE Test Pattern Aggressive Execution Time Optimization

Shen Lim Te, Hin Fook Wong, James Chee Liang Teoh (Intel Corporation, Malaysia)

Abstract— As semiconductor device complexity continues to grow, the cost associated with High Volume Manufacturing (HVM) testing becomes a significant bottleneck. This paper introduces a TRIZ-inspired optimization methodology and strategy aimed at maximizing Automated Test Equipment (ATE) test patterns wait cycle efficiency during test pattern execution to achieve aggressive cost savings.

14. Complex Timing Exceptions Handling in Transition Mode ATPG in Automotive SoCs

Tripti Gupta, Shalini Pathak, Monica Sharma and Prince Kumar (STMicroelectronics Ltd. India)

Abstract— Automotive SoCs have a very stringent requirement in terms of stuck_at and transition/at_speed test coverage. Typical value of transition coverage is 90%. Achieving this coverage becomes all the more challenging if functional timing exceptions are too many to handle in at_speed ATPG because most of the constraints which are required in functional mode of the device are also applicable for at_speed ATPG. Depending on the architecture and application of the automotive SoC, there could be complex timing exceptions like various multi cycle paths (MCPs) and false paths. This makes the design constraints file (SDC) very bulky at times. The Static Timing Analysis (STA) team takes this file and does the timing closure across corners in functional mode. The challenge lies with at_speed ATPG mode where it becomes difficult to use the same SDC file as it is because most of the ATPG tools cannot understand all the commands used in this kind of file and hence need a processing on this file before it can be used to generate patterns. The poster describes the solution adopted in one of the automotive MCU where the SDC file was too heavy for the ATPG tool to even read and then comprehend all the exceptions to be applied during pattern generation. There were hundreds of MCPs of various degrees like MCP of 2,4,8 etc. Mostly in such cases, MCPs are treated as false paths and we lose all the test coverage on such paths. So, we tried to manage the volume of SDC file and MCPs through Synopsys violation based exception file

POSTER SESSION

writing using PT2TMAX flow. As such the flow was already existing but we exploited it in a way to gain maximum advantage in test coverage from MCPs also. We devised the following steps to derive multiple SDCs through this flow and generated patterns for both false paths and MCPs effectively: 1. Variables/parameters were added in functional SDC to enable multiple violation based SDCs to be given to ATPG tool. For Eg : paths valid in functional STA but not valid for ATPG, paths not visible in STA during ATPG configuration. 2. MCPs were segregated based on their clock relations like MCPs between synchronous domains or MCPs within the same clock domain. Synchronous OCC waveforms were leveraged to target the paths that can be captured at the fastest frequency or the slowest frequency. 3. Violations were dumped in different sessions based on the parameter settings 4. Exceptions were written using `write_exceptions_from_violations` command across timing corners 5. Patterns were generated using violation based SDCs in multiple passes and test coverage could be achieved from various categories of MCPs also. There are definitely improvements that can be incorporated in our EDA tools to handle such scenarios but the flow that we used can actually serve as a savior till the time we do not have a robust way of managing functional SDCs in ATPG environment.

15. SLM: Coverage Enhancement through a Robust Stitching Algorithm with Path Margin Unit (PMU) Count and Design Traits

Mohit Verma, Veeramani Rajalingam and Uravi Saraogi (Synopsys, India)

Abstract— Monitoring, Analyzing, and Optimizing health of silicon is possible through Silicon Lifecycle Management (SLM) IP. This can be achieved by putting sensors like Path Monitor Unit (PMU) into designs which will monitor the design instances through timing endpoints. Monitoring effectiveness is gauged by the percentage of design instances covered by PMUs. Two designs were analyzed, showing distinct coverage despite using the same stitching algorithm and calculated PMU count. The design with lower coverage underwent trials with varying PMU counts, resulting in improved coverage through a robust flow. Limitations related to design nature were discussed in detail, highlighting differences compared to the other design.

16. Testing of Retention capable GPIOs through Boundary Scan

Avik Sen, Maheedhar Jalasutram and Ranjit Kumar Kurra (Google India Bangalore, India)

Abstract— In Mobile applications, retaining the IO state is essential, so that the device wakes up faster post sleep or hibernate state. This study explores the method used for testing the Retention capability for GPIOs using the 1149.1 JTAG standard. Existing JTAG BSCAN test flow is utilized to test for any static faults on the Retention pin of the GPIOs.

17. A Comprehensive Methodology for Early Validation of Timing Constraints

Sravan Kumar, Varsha M, Anjali Konakanchi, Meghana L, Leela Krishna Thota (Synopsys India Pvt Ltd, India)

Abstract— Timing constraints are used to specify the timing characteristics of a design. Small errors in these constraints may cause a significant impact on internal timing interconnections, flip-flops, delays

POSTER SESSION

through logic, etc. Traditionally, timing constraint validation used to occur post-synthesis leading to a higher turnaround time for constraint corrections. So, it is essential to verify timing constraints in the early stages of the design flow. Hence, this paper presents a comprehensive methodology leveraging the tool for constraint management (TCM) to help designers validate timing constraints directly on RTL. The proposed methodology enhances the validation of timing constraints, clock domain crossing (CDC) verification, false path (FP) verification, and assertions in the RTL stage. The CDC verification has been done by generating assertion files during static analysis, ensuring signals crossing different clock domains meet timing requirements. The FP verification has been addressed by capturing exclusion criteria for specific paths, thereby preventing erroneous timing analysis results, and optimizing design performance. This methodology also encompasses assertion generation to validate design behaviors, providing a comprehensive approach to early-stage design validation. Overall, this approach enhances design robustness and reliability, particularly in complex SoCs with multiple clock domains.

18. A Scalable Approach to Energy Management Features Validation

Mehul Kumar Patel, Adarsh C S and Niharika Misra (IBM, India)

Abstract— With the growing industry emphasis on sustainability, energy efficiency of computer servers is gaining huge importance. Due to its inherent design, validating energy management features in modern computer servers is considered very critical & complex task. Effective energy management relies on seamless coordination between hardware, firmware, and operating systems, making the validation process inherently complex. This poster presents the tools and methodology developed for validating power management features in IBM Power systems. It outlines an automated workflow that configures the system, orchestrates representative workloads, performs closed-loop validation of energy management functions, and generates comprehensive reports and logs. By ensuring accurate validation of power-saving features, the approach contributes to building energy-efficient systems that align with broader sustainability goals, including reduced data center energy consumption and environmental impact.

19. Automation to Speed up the process of Timing Closure of the JTAG Network

Kshitij Kulshreshtha and Amihay Rabenu (SYNOPSYS Inc, India) , Manish Arora (SYNOPSYS Inc, United States)

Abstract— IEEE 1687 has been implemented to qualify a significantly sophisticated, compliant, and automated access to the mounting number of embedded instruments in today's integrated circuits and the vital constituent of 1687 Network is the SIB (Segment Insertion Bit) and the Enhanced TAP Network. This Enhanced TAP Network is having the notable gated signals For example, ShiftEn , CaptureEn , UpdateEn and Select. Jtag CSU control signals are shared across all the jtag network this will lead to a major obstacle in doing the timing closer since long distances are slowing down the Cycles's frequency. Test time become very long due to Low frequency. The FSM of the TAP with which the whole system works is very critical and if the state transitions are not ensuing appropriately then the 1687 network is not going to work properly. The timing of control signals ShiftEn, CaptureEn and UpdateEn is essential and can create problem during timing closure activity also it should not impact the reverse log mapping diagnosis flow. This reverse Log mapping diagnosis flow is also critical since it can hamper the User for right Test cycles and for the calculation of right offset for the Test cycles. Motivation is to create an ecosystem to alleviate both Timings obstacles and Test-Time reduction.

POSTER SESSION

20. Enhanced Security Mechanism for JTAG Network

Kshitij Kulshreshtha and Vistrita Tyagi (SYNOPTYS Inc, India), Manish Arora (SYNOPTYS Inc, United States)

Abstract— Chips are becoming multifaceted day by day and they hold mammoth amount of affluence in the form of Embedded Instruments, hi-end Sensors, hardware monitors, BIST Engines, OCC Controllers etc. To protect this unabridged ecosystem, Encryption and Obfuscation practices need be espoused. Since we are obligated to use the Test and debug circuitry for testing the DFT logic, it regrettably opens the door for an unauthorized admittance to Internal registers of Instruments which can lead to reverse engineering. It results in counterfeiting and data theft attacks which is a very protuberant security challenges which would create a huge risk for the device itself. We are proposing a distinctive mechanism to secure the 1687 Network by embracing a counter strategy to thwart this attack to appease below stated challenges. Securing the 1687 Network by adopting a small, scaled Hardware in the design. Selectively locking the SIBs to secure the Highly secure Instruments in the design.

21. Optimizing Mobile & Automotive GPUs with Streaming Fabric, SEQ/XLBIST, IEEE1687 and TSO.ai

Shamitha Rao and Deepika Reddy Yenna (SYNOPTYS Inc, India)

Abstract— This poster demonstrates how Manufacturing and In-system Test requirements of Mobile and Automotive GPUs are met with Streaming Fabric, SEQ/XLBIST unified codec and 1687. It also showcases Automatic Test Pattern Generation (ATPG) cycle reductions using TSO.ai and SEQ diagnosability thus optimizing implementation turnaround time, test cost and test quality.

22. Resurgence in Advanced ATPG Techniques for High-Performance Designs

Dale Meehl, Brion Keller, Krishna Chakravadhanula and Pradeep Nagaraj (Cadence Design Systems San Jose, CA, USA)

Abstract— The current era of high-performance designs bring with them challenges in ensuring high quality Test application. As designs grow in size and complexity, the DFT (Design For Test) and ATPG (Automatic Test Pattern Generation) capabilities have to adapt for functional data paths being tuned for high speeds, which is leading to growing trend in having non-scan flops and pipelining in the design. Advanced ATPG techniques can help address such partial-scan designs while continuing to achieve high fault coverage with compact pattern counts. In this poster, we propose couple key technologies customers can enable to minimize scan logic but keep ATPG quality.

Test Reality Check (TRC)

Session Chair : Rajit Karmakar

TRC 1.1: IDDq Vector Quality (FTRV) Improvement on Mature Technodes through Prime Time Power EXTension (PTPX) Simulation at Pre-Silicon

Marijune Samaniego, Hariharan Krishnamurthy (Qualcomm, Singapore); Sudheer Yadapalli, Shilpa Bhardwaj, Tanu Pruthi and Maharshi Pandya (Qualcomm, India)

TRC 1.2: . Incremental Memory Repair Techniques with Synopsys SMS for Multi-hierarchy Server Architecture

Ankit Bansal (Synopsys, India); and Ajay Chauhan (Marvell, India)

TRC 1.3: . Automated and Verifiable Security Exclusion Management for Scan-Based Debug: The SDEX Methodology

Krunal Siddhapathak and Harsharaj Ellur (Google, India)

TRC 1.4: Redefining Low Power Scan Testing: A Novel Asymmetric Clock Gating

Arshdeep Singh, Navaneeth R and Vishal Diwan (Texas Instruments, India)

TRC 1.5: Breaking the Drop: Innovative Methods for IR Drop Mitigation in Scan Testing

Rahul Pandey, Karthik Potturu, Suresh Ambati and Dhirendra Sharma (Samsung, India)

Learning Corner Series

Session Chair : Veejaye Panayadiyan

Learning Corner Series – ATMP/OSAT Fundamentals

ATE in Action: Teradyne on Semiconductor Manufacturing Test & Value Creation



Paresh Bharkhada
Managing Director
Teradyne

Paresh Bharkhada is a seasoned Managing Director at Teradyne, leading strategic growth, investments, and the expansion of the company's footprint in India and Emerging Markets. With over 20 years of diverse experience in the semiconductor industry, he has excelled in roles spanning engineering, management, and business development. Prior to joining Teradyne, Paresh played a pivotal role at Intel, where he successfully spearheaded and scaled Intel's Development Cloud initiative, driving significant innovation and business impact. His strong technical background, combined with business acumen & strategic leadership, allows him to create sustainable growth and technological advancements. Paresh holds a Master of Business Administration from the prestigious Haas School of Business and an Master of Science in Electrical Engineering from the University of Windsor, Canada, positioning him as a dynamic leader in technology and business strategy.

Scaling with Confidence: Seamless Navigation for High-Volume Semiconductor Testing



Mr. Larry Liow
Director,
Sales and Marketing,
Advantest Singapore

Larry has more than 25 years of experience in the semiconductor test business. His role includes strategizing corporate roadmap, industry engagements, business development and new product introduction in Southeast Asia and India region. He has held multiple portfolios including Sales and Marketing, Customer Support, Business Support Administration and System Application Engineering.

Larry holds a Master of Business Administration from the University of South Australia and Bachelor of Engineering Management from University of Western Sydney. He leads the vision to bridge the gap between industry and academia, providing leadership and guidance in connecting academia with Advantest, to enable talent and skill developments for the electronic industry.

Learning Corner Series

Session Chair : Veejaye Panayadiyan

Learning Corner Series – ATMP/OSAT Fundamentals

Production Infrastructure and ATE Environment - The Fundamentals, Challenges of Existing Flows and Future Trends



Gaurav Mittal
SMTS,
Texas Instruments

Gaurav has 17+ years of industry experience in analog test engineering. He is currently senior member of technical staff (SMTS) and the test and validation manager for the Multiphase Processor Power group where he focuses on devices for Enterprise market (Server, AI, Data Centers). Previously he has been the architect and manager for Power Switching, Interface and Lighting business units at Texas Instruments, owning power protection and distribution devices across all analog market segments including automotive, industrial, PE, communication, and enterprise. He has defined the test roadmap strategy, identifying the test gaps, and driving the systematic development flow to have self-sustaining test solutions. Since 2008, he has worked and defined test strategies across a variety of power devices including eFuses, Hot-swap, Multiphase controllers and power stages, LDO, ideal diode, current loop protectors, and high side drivers. He has 18 papers (external and internal), 1 patent, 8 industry talk at different prestigious conference.



Rajesh Jabade
MGTS, MPP,
Texas Instruments

Rajesh is currently Member group of technical staff (MGTS) and Test manager in multiphase processor power solutions team in TI. He joined TI in 2016 and from then instrumental in leading multiple power teams in Test and characterization domain. Prior joining TI he has worked with Nvidia Graphics and KarMic Design as a test and characterization engineer. He has vast experience in developing Analog, Digital and Mixed signal testing on various tester platforms. Rajesh has master's degree in Microelectronics from BITS Pilani.

HACKATHON

VLSI TEST COVERAGE OPTIMIZATION HACKATHON

The hackathon comprised two rounds, with the first serving as a qualifier. Participants were challenged with real-world RTL design and validation problems, including scan DRC analysis, test coverage optimization, and fault coverage enhancement. The hackathon emphasized teamwork, technical proficiency, and effective time management, offering a practical, hands-on experience in RTL test and debug processes.

Ask ChatGPT

Winners

Team Name	Institute	Members
Titans	KLE Technological University	Jagadeesh Pradhani, Suresh G Kini, L Ashok Kumar Reddy
Flip-Flop Fixers	BMS College of Engineering	Chaitra YG, Hady Rahman AV
Quantum Bytes	IIIT Guwahati	Dr. Babita Jajodia, Bheemuni Harshavardhan Reddy, K. Shruthi

Honourable Mention

Team Name	Company	Members
Impro_DFT	Einfochips (An Arrow Company) Pvt. Ltd.	Ila Vaghela, Amansinh M Chudasama, Aashvi Patel